

Static Gesture Classification — Run Report

Generated 2026-05-13 11:46:54

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 17	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	17 · 559 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	78 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 4212
Split / seed / CV	random_state=42 · CV: LOUO (16 folds, every user held out once)
Sample counts	train (post-aug)=559 · LOUO-evaluated samples=559

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 16 folds)	CV std	CV min	CV max
SVM (Linear)	0.8031	0.1035	0.6216	0.9714
Stacking (SVM+RF+KNN+LogReg)	0.8012	0.1109	0.5946	0.9714
Voting Soft (SVM+RF+KNN+LogReg)	0.7998	0.1030	0.5405	0.9714
Logistic Regression	0.7922	0.1158	0.5135	0.9714
SVM (RBF)	0.7904	0.0954	0.5946	0.9429
Random Forest	0.7701	0.1410	0.4571	0.9714
LDA	0.7473	0.0974	0.5135	0.8857
KNN	0.6179	0.0804	0.4722	0.7714

Per-class CV F1 (across all LOUO folds)

Class	SVM (Linear)	Stacking	Voting (soft)	Logistic Regression	SVM (RBF)	Random Forest	LDA	KNN	Support
Double_Closed_Fist	0.742	0.695	0.727	0.713	0.696	0.644	0.671	0.536	83
Double_Nothing	0.690	0.694	0.667	0.667	0.681	0.580	0.602	0.268	75
Double_Okay	0.855	0.929	0.895	0.861	0.863	0.892	0.789	0.509	80
Double_Open_Palm	0.829	0.867	0.842	0.813	0.834	0.889	0.763	0.659	80
Double_Phone	0.636	0.552	0.587	0.611	0.585	0.465	0.535	0.364	80

Double_Pistol	0.906	0.924	0.914	0.931	0.930	0.902	0.902	0.905	81
Double_Spiderman	0.920	0.892	0.893	0.888	0.892	0.918	0.906	0.852	80
macro avg	0.797	0.793	0.789	0.783	0.783	0.756	0.738	0.585	559

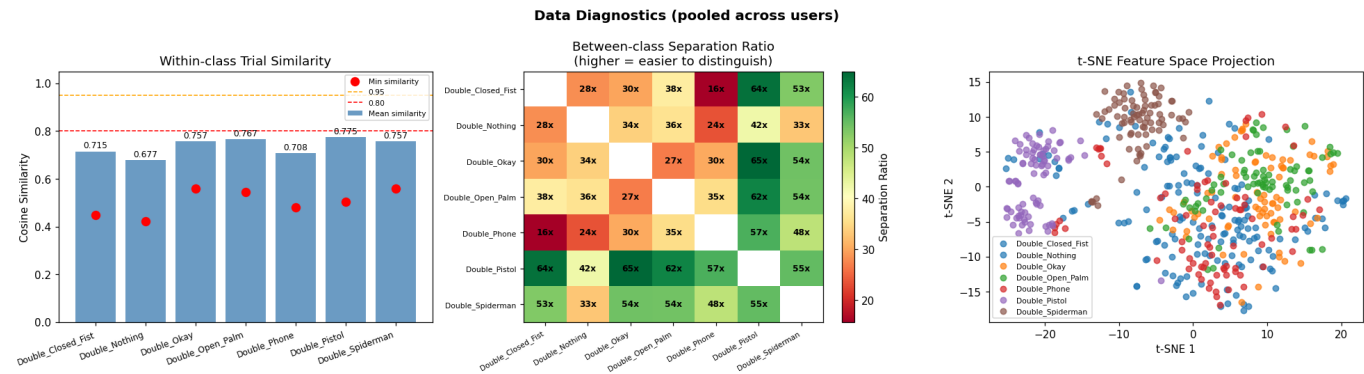
Per-user accuracy (each user as held-out fold)

User	n trials	SVM (Linear)	Stacking	Voting (soft)	Logistic Regression	SVM (RBF)	Random Forest	LDA	KNN
1_Alan	35	0.771	0.857	0.857	0.829	0.829	0.829	0.743	0.486
2_Alex	35	0.629	0.686	0.657	0.600	0.629	0.457	0.629	0.686
3_Anghad	35	0.829	0.829	0.829	0.800	0.771	0.829	0.743	0.600
4_Daniel	35	0.800	0.800	0.743	0.771	0.771	0.857	0.829	0.600
5_Georgia	35	0.800	0.657	0.771	0.743	0.714	0.571	0.886	0.686
6_Harry	35	0.971	0.971	0.971	0.971	0.943	0.971	0.829	0.771
7_Henry	35	0.629	0.629	0.743	0.686	0.743	0.629	0.571	0.657
8_Jestin	36	0.778	0.722	0.694	0.722	0.694	0.833	0.722	0.472
9_Joselyn	30	0.933	0.900	0.900	0.867	0.867	0.833	0.800	0.633
10_Josh	35	0.857	0.829	0.829	0.886	0.829	0.714	0.743	0.514
11_Mansh	36	0.917	0.944	0.861	0.944	0.833	0.861	0.778	0.611
12_Marcus	35	0.800	0.857	0.829	0.800	0.857	0.971	0.771	0.657
13_Nat	37	0.622	0.595	0.541	0.514	0.595	0.622	0.514	0.541
14_StephenV2	35	0.771	0.857	0.886	0.857	0.829	0.657	0.857	0.657
16_Bella	35	0.829	0.771	0.800	0.829	0.800	0.857	0.800	0.600
17_Raquel	35	0.914	0.914	0.886	0.857	0.943	0.829	0.743	0.714
Mean (across users)		0.803	0.801	0.800	0.792	0.790	0.770	0.747	0.618
Std (across users)		0.103	0.111	0.103	0.116	0.095	0.141	0.097	0.080

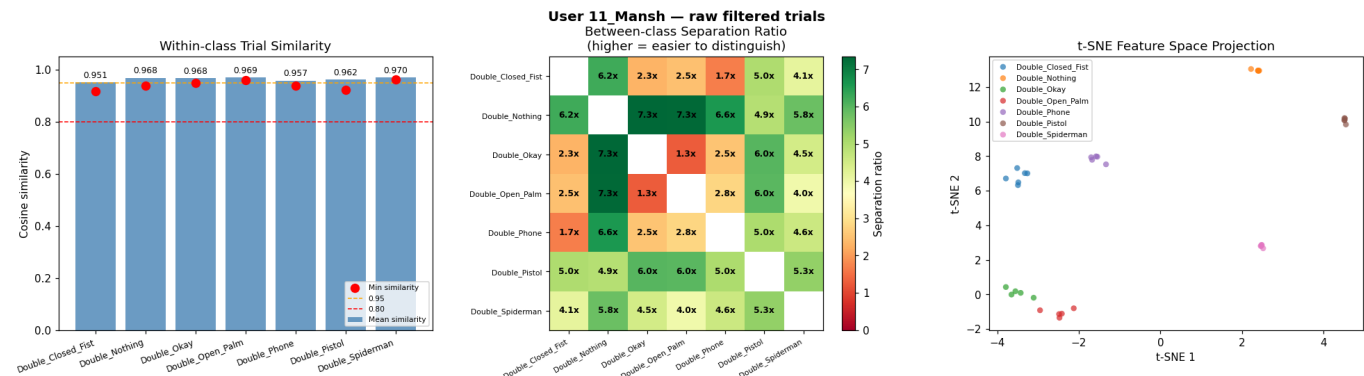
Appendix — Data Diagnostics

75 figures: pooled across users + one per detected user (raw filtered trials and extracted-feature views).

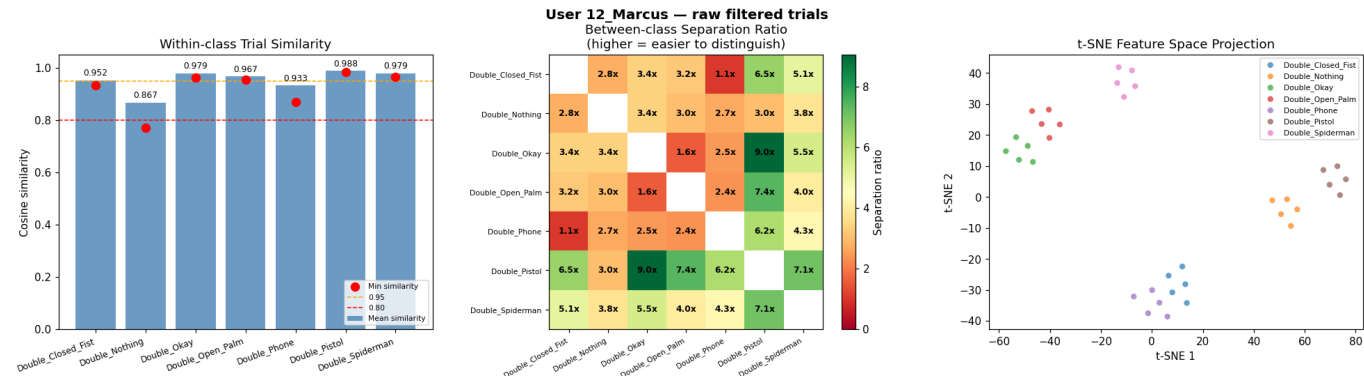
Pooled diagnostics — extracted features (across all users)



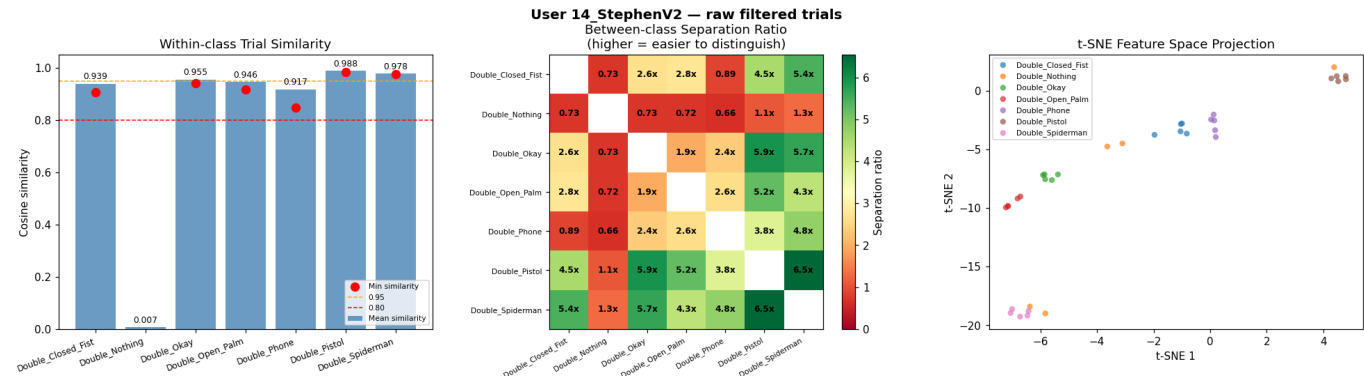
User 11_Mansh — raw filtered trials



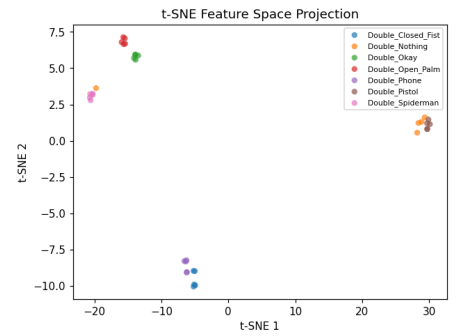
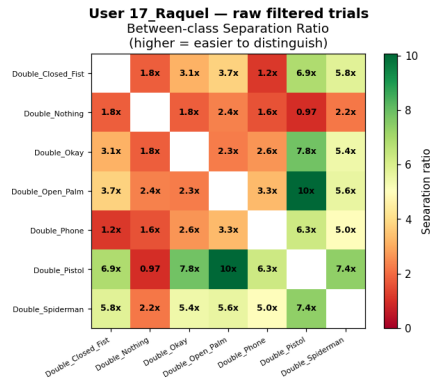
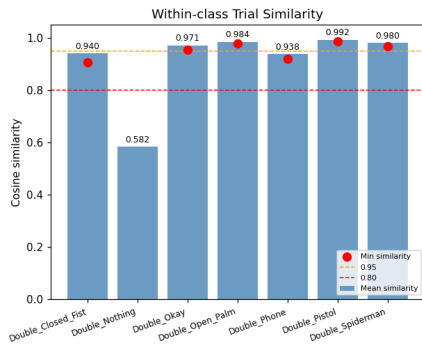
User 12_Marcus — raw filtered trials



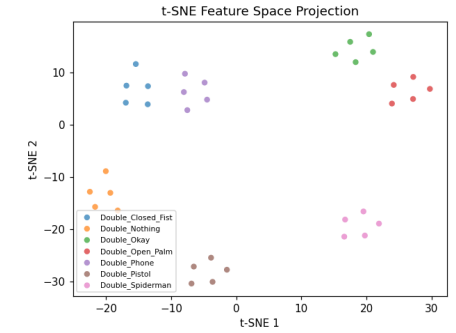
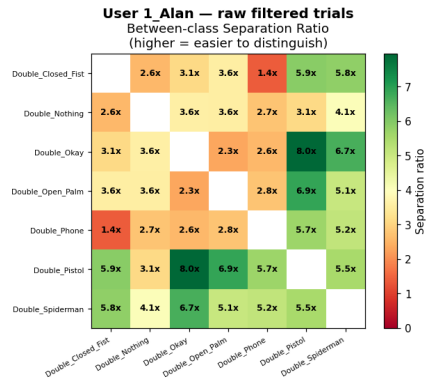
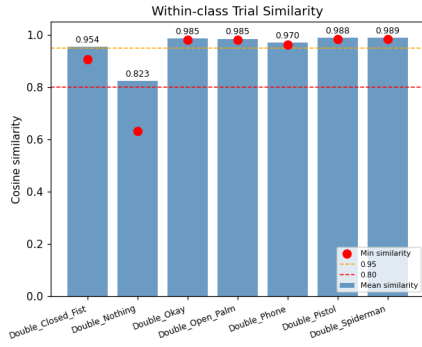
User 14_StephenV2 — raw filtered trials



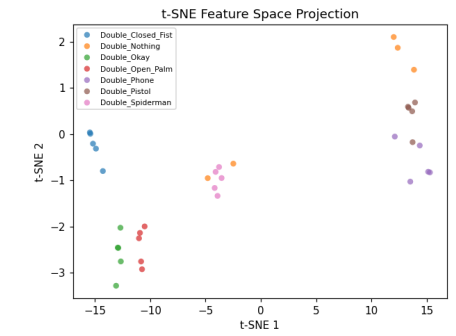
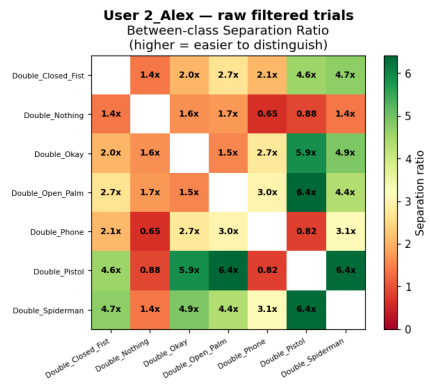
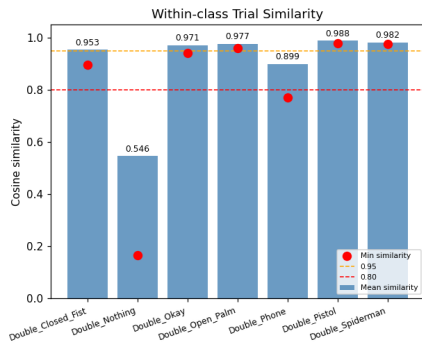
User 17_Raquel — raw filtered trials



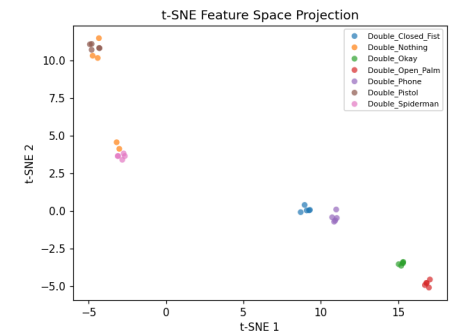
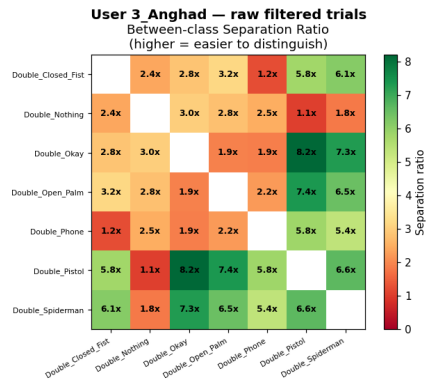
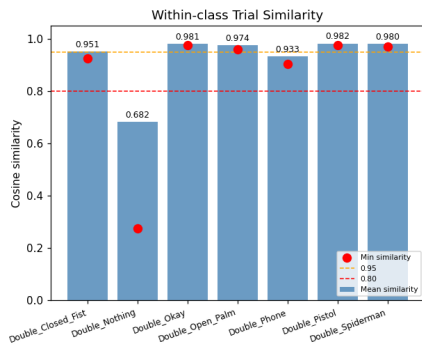
User 1 Alan — raw filtered trials



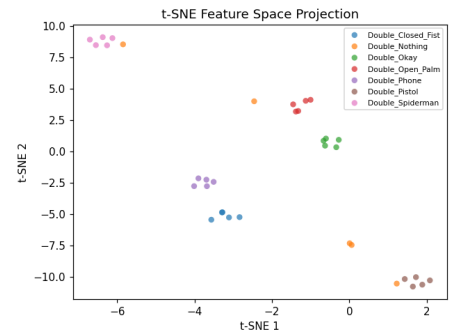
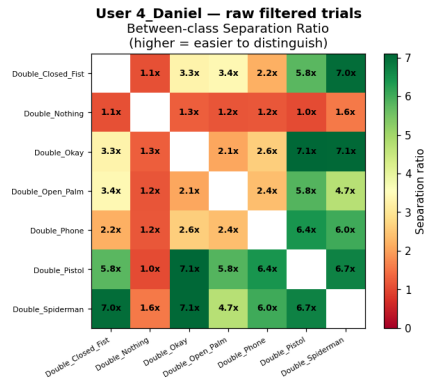
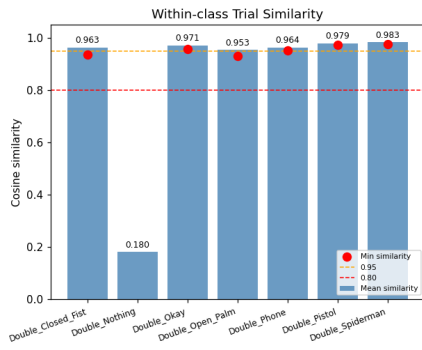
User 2 Alex — raw filtered trials



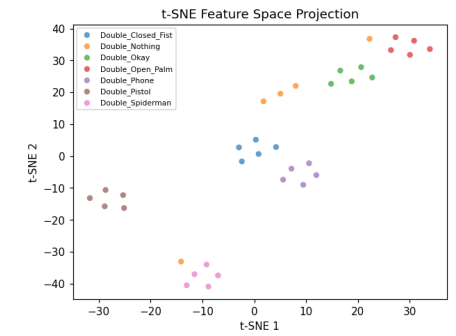
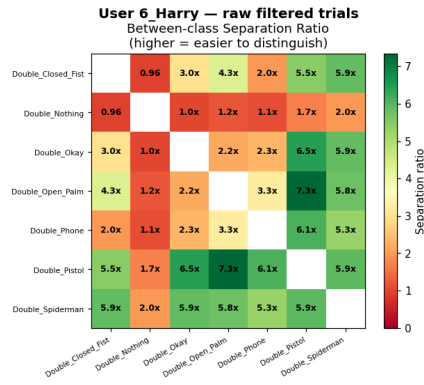
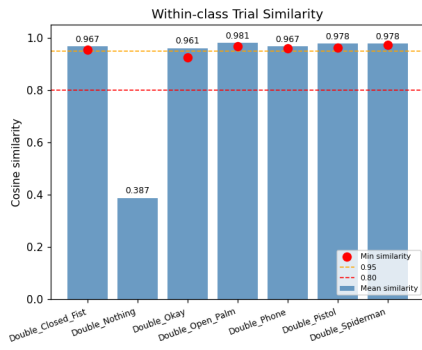
User 3 Anghad — raw filtered trials



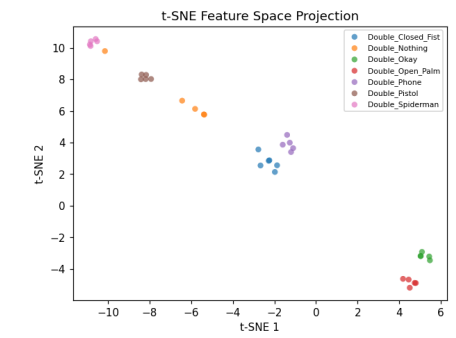
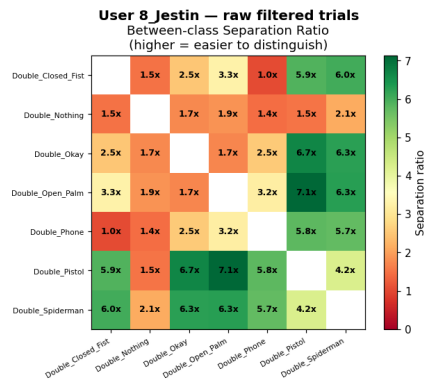
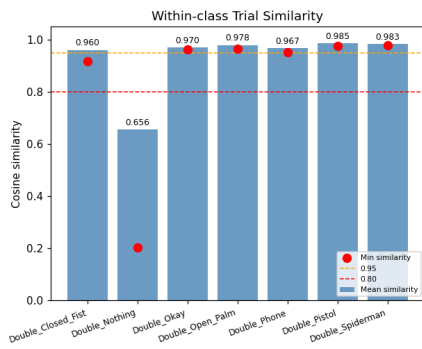
User 4 Daniel — raw filtered trials



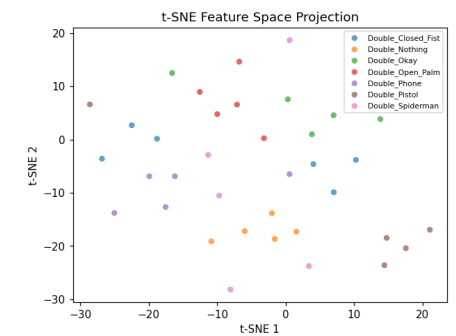
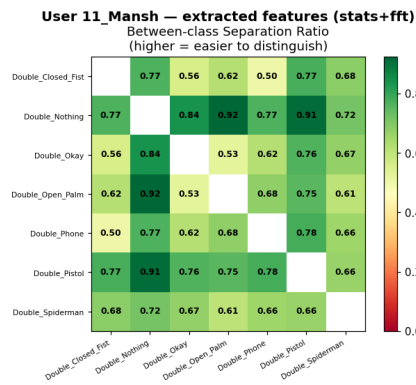
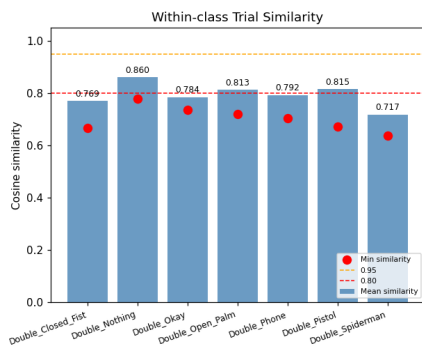
User 6_Harry — raw filtered trials



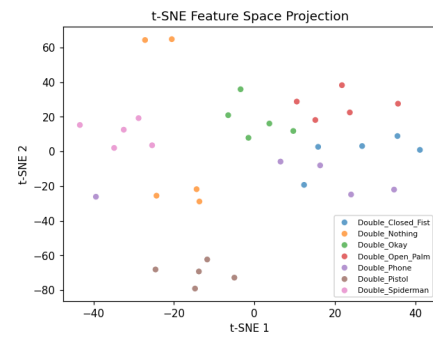
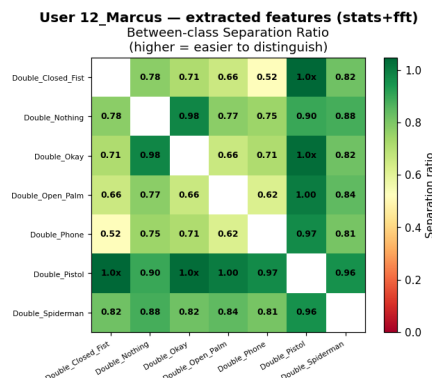
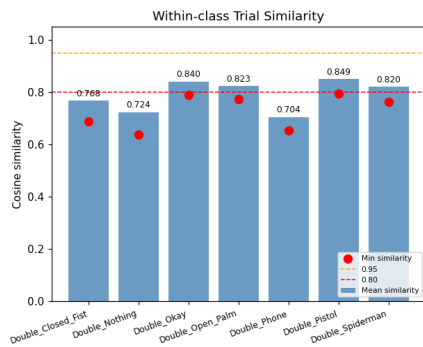
User 8_Jestin — raw filtered trials



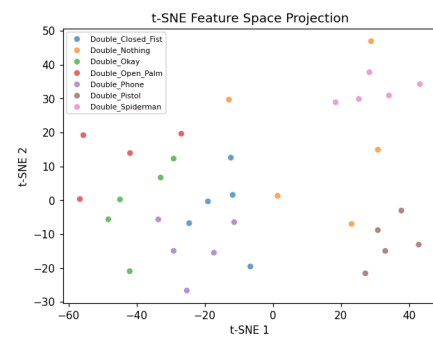
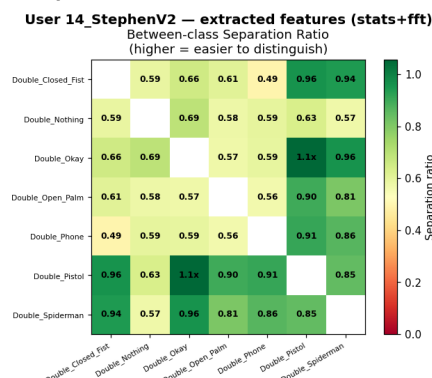
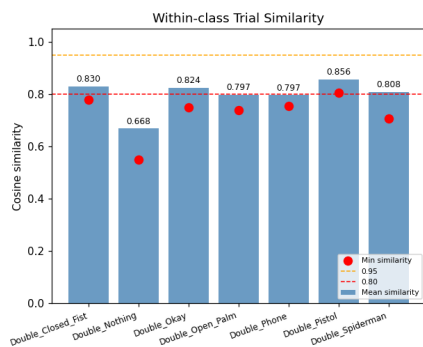
User 11_Mansh — extracted features (stats+fft)



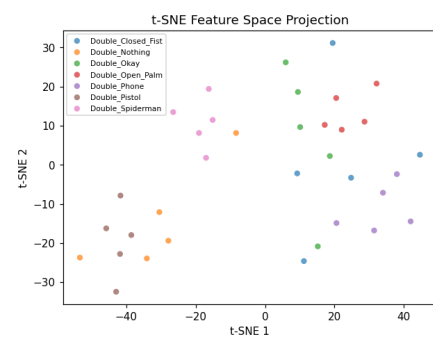
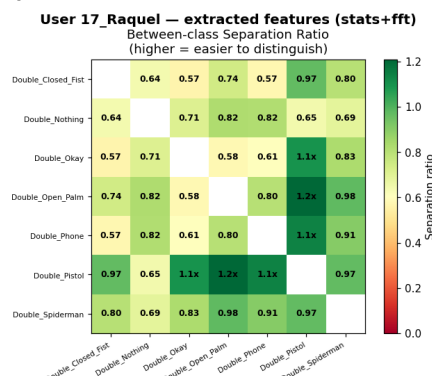
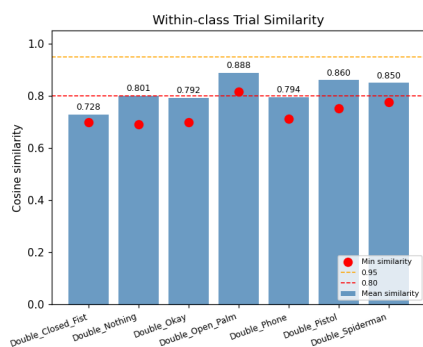
User 12_Marcus — extracted features (stats+fft)



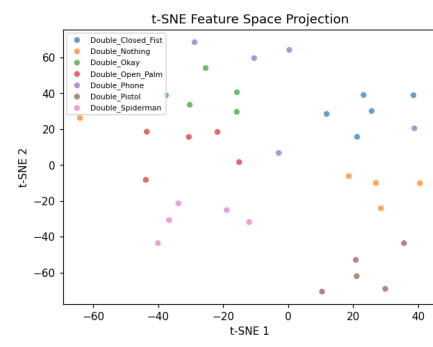
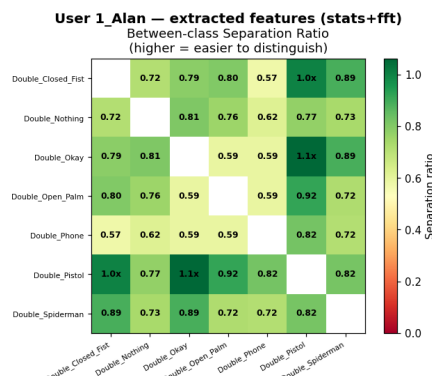
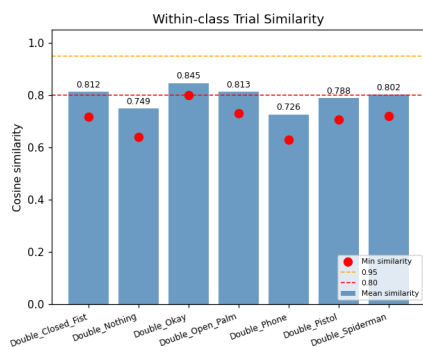
User 14_StephenV2 — extracted features (stats+fft)



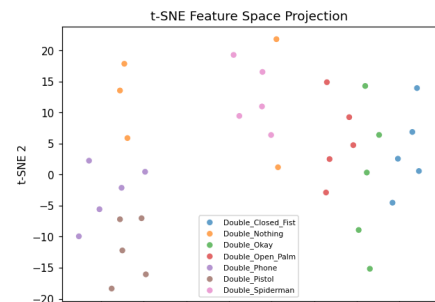
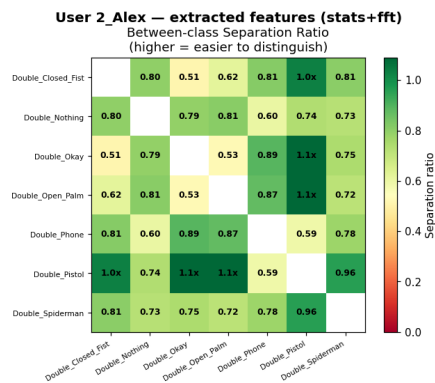
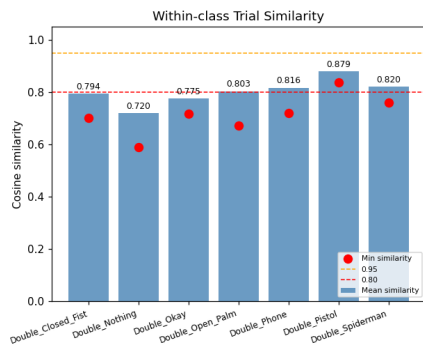
User 17_Raquel — extracted features (stats+fft)



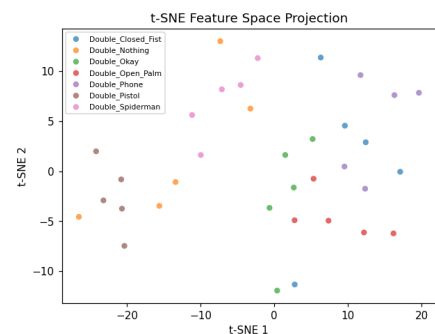
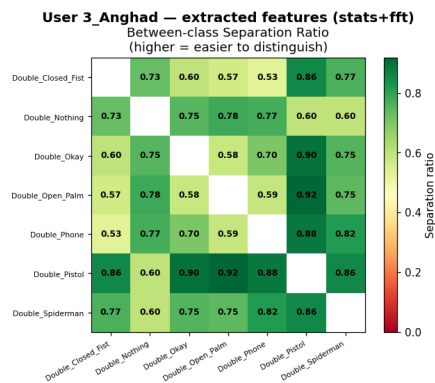
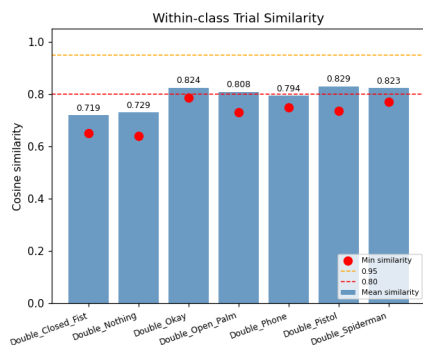
User 1_Alan — extracted features (stats+fft)



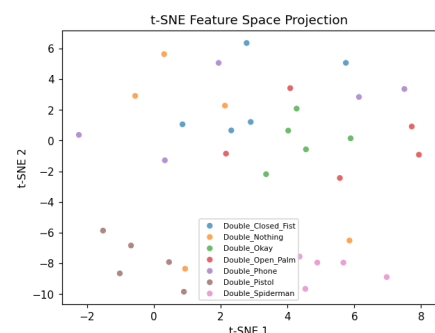
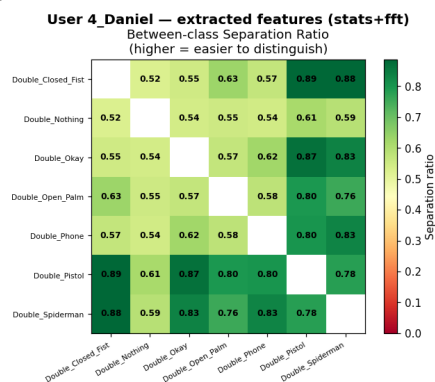
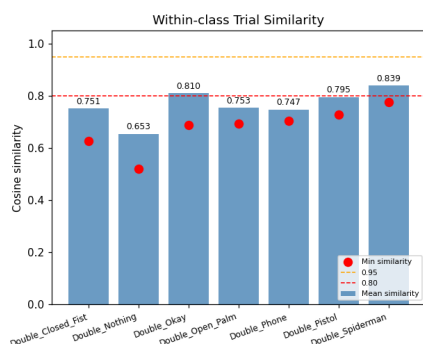
User 2_Alex — extracted features (stats+fft)



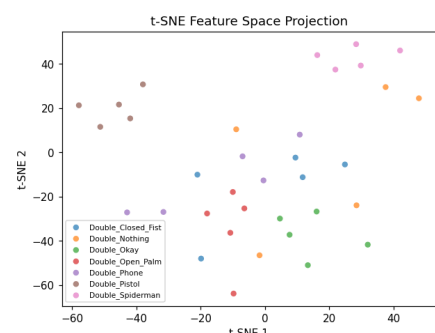
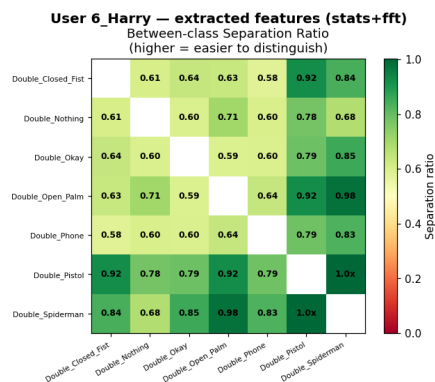
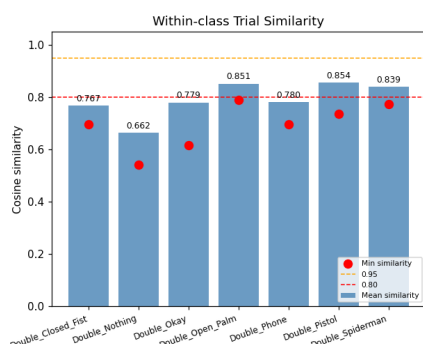
User 3_Anghad — extracted features (stats+fft)



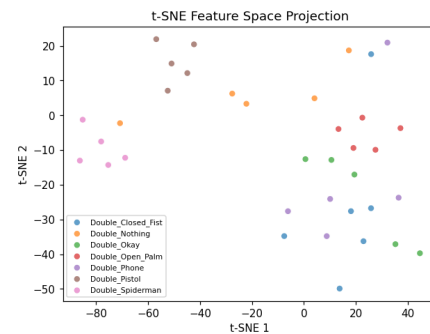
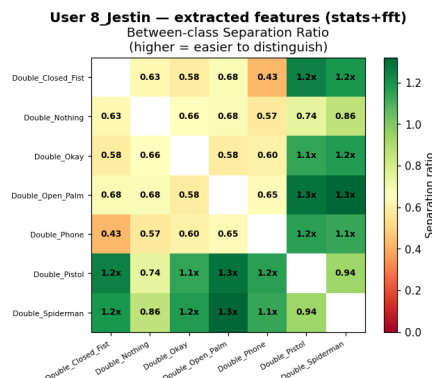
User 4_Daniel — extracted features (stats+fft)



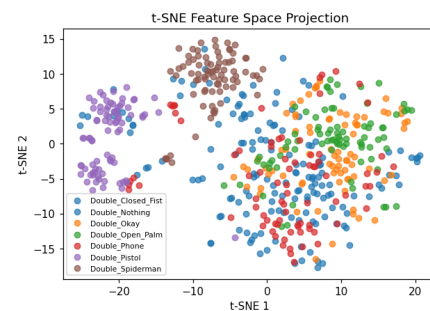
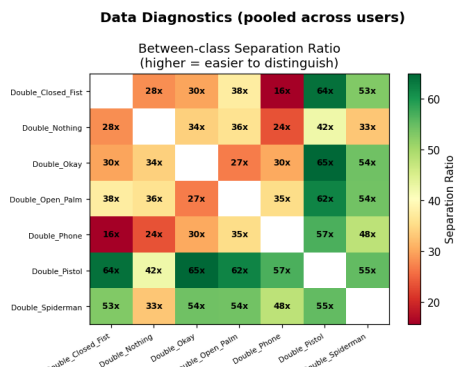
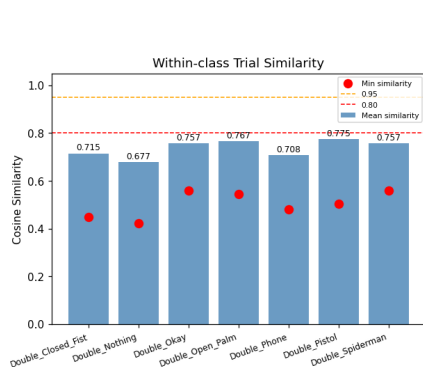
User 6_Harry — extracted features (stats+fft)



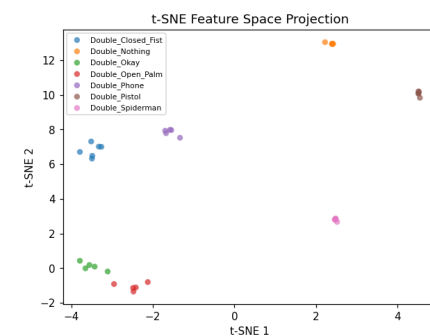
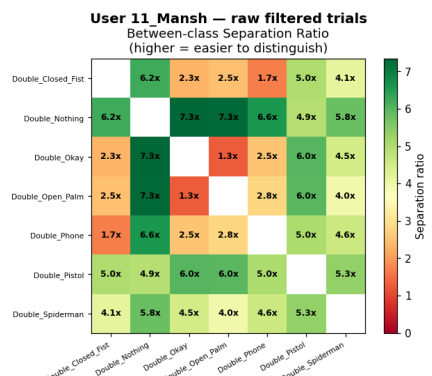
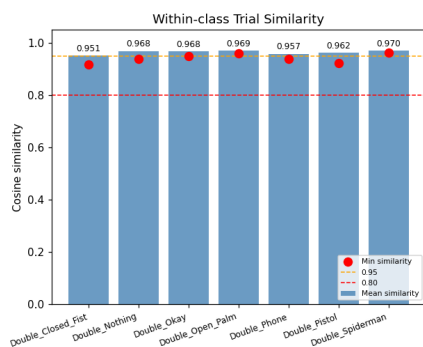
User 8_Jestin — extracted features (stats+fft)



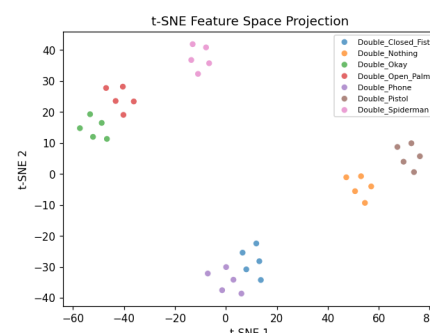
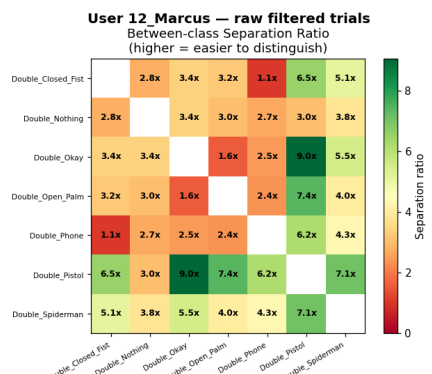
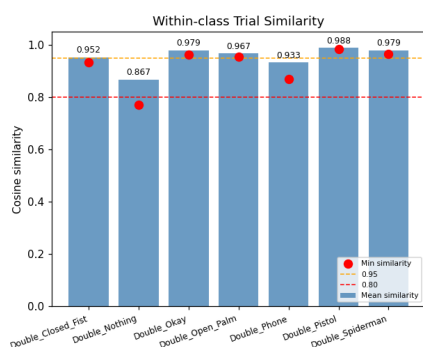
Pooled diagnostics — extracted features (across all users)



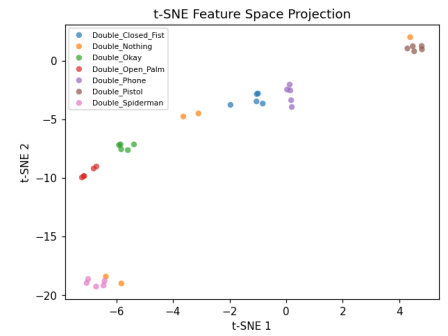
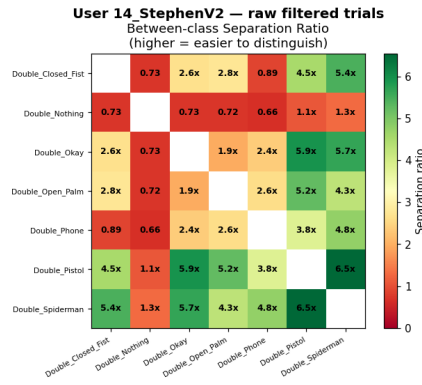
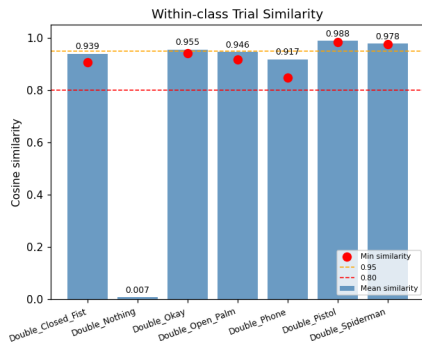
User 11_Mansh — raw filtered trials



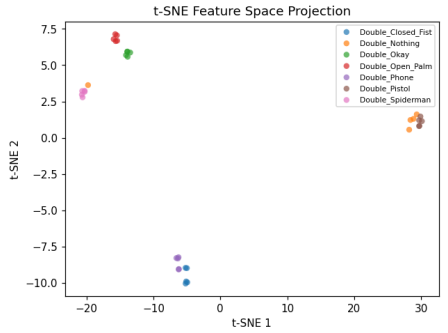
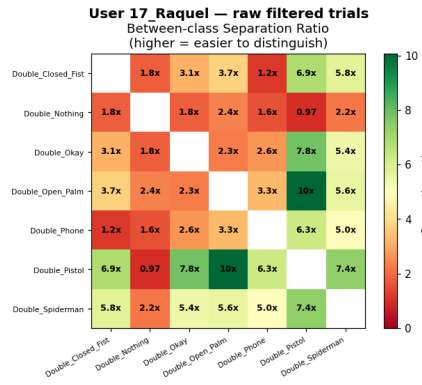
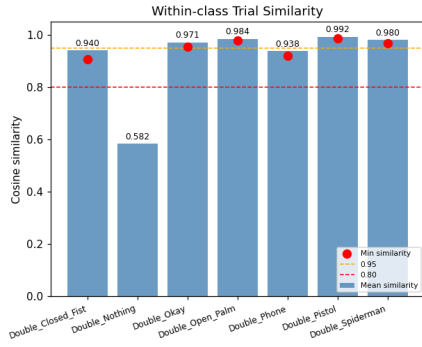
User 12_Marcus — raw filtered trials



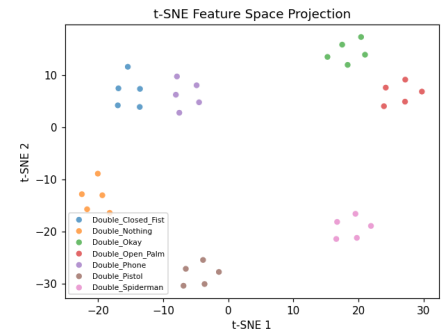
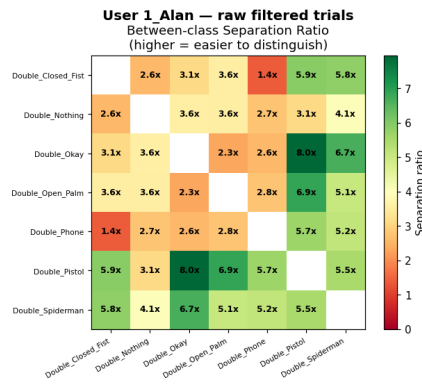
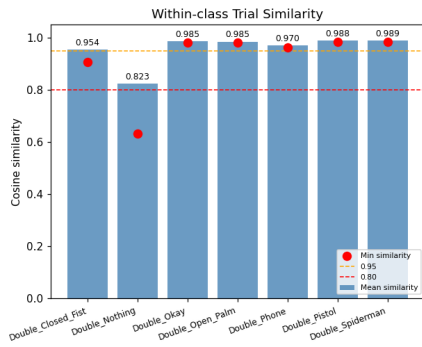
User 14_StephenV2 — raw filtered trials



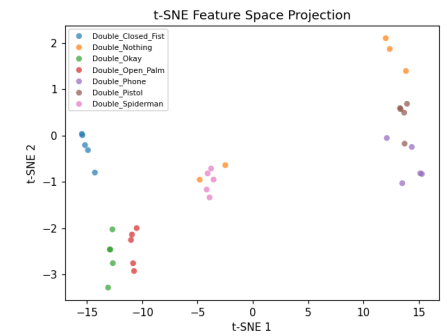
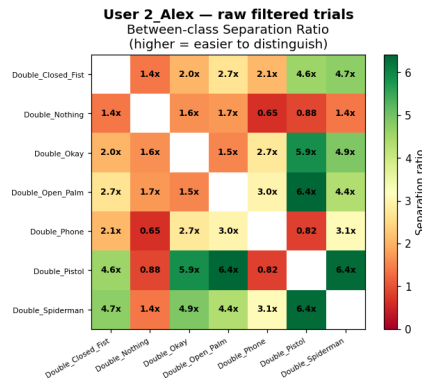
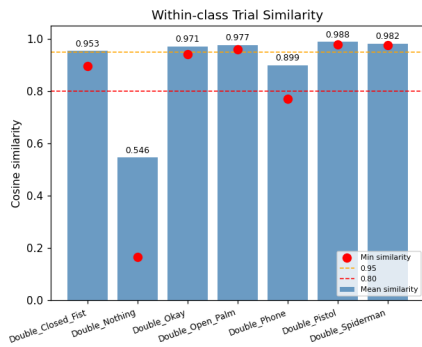
User 17_Raquel — raw filtered trials



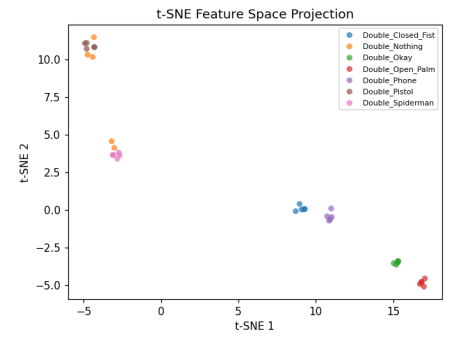
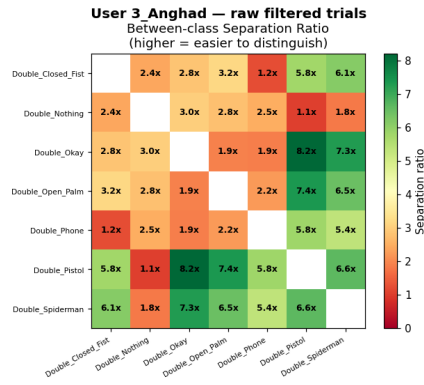
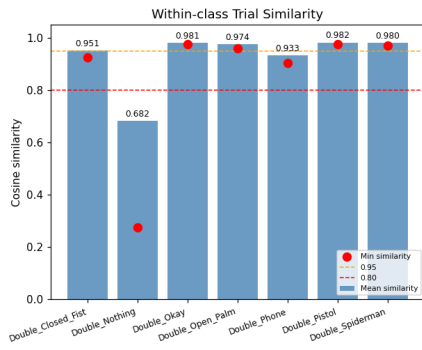
User 1_Alan — raw filtered trials



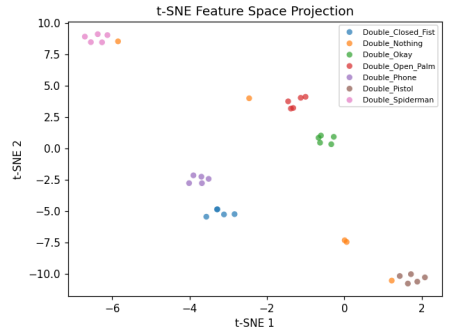
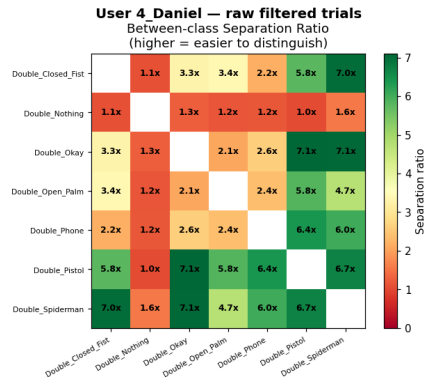
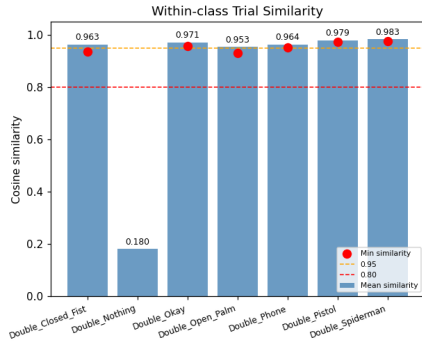
User 2_Alex — raw filtered trials



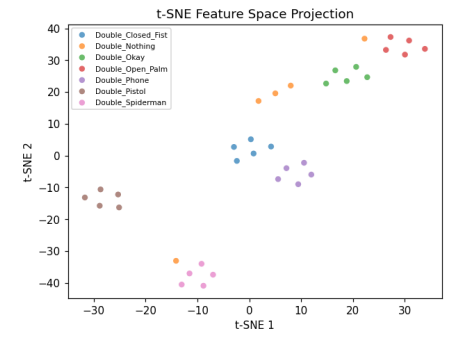
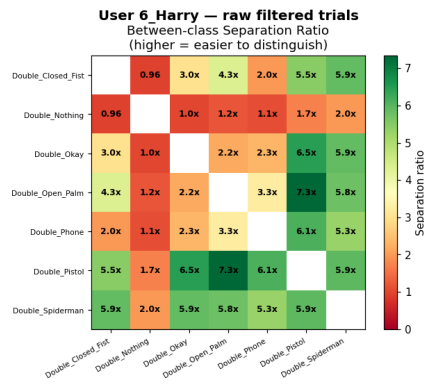
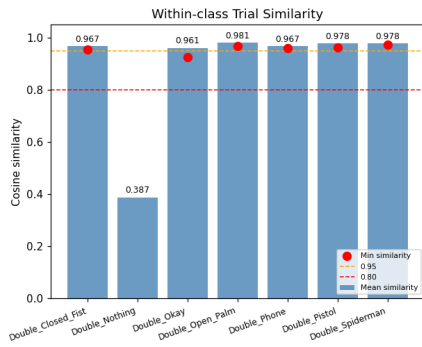
User 3_Angbad — raw filtered trials



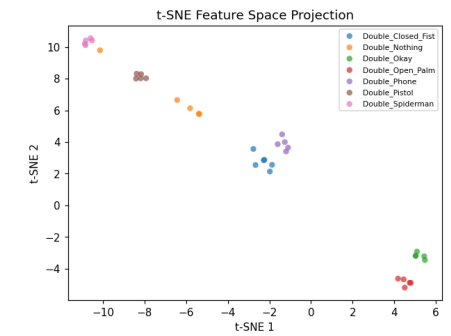
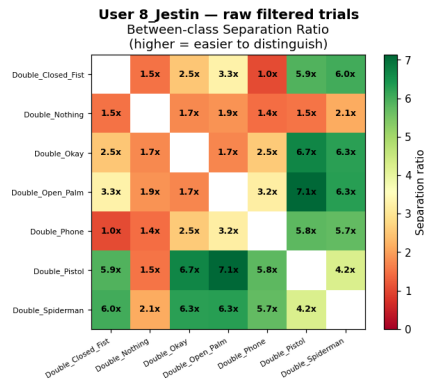
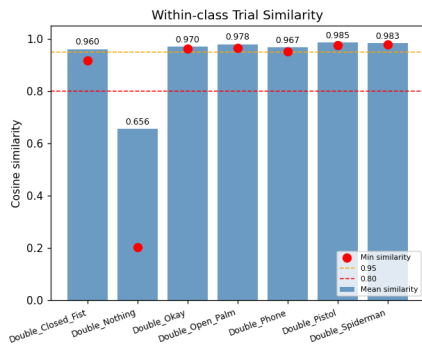
User 4_Daniel — raw filtered trials



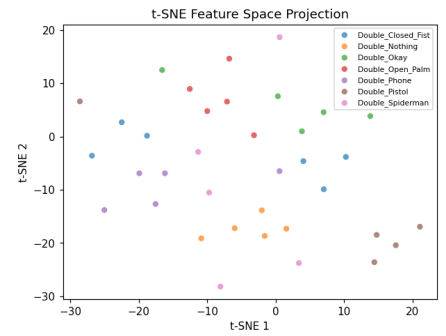
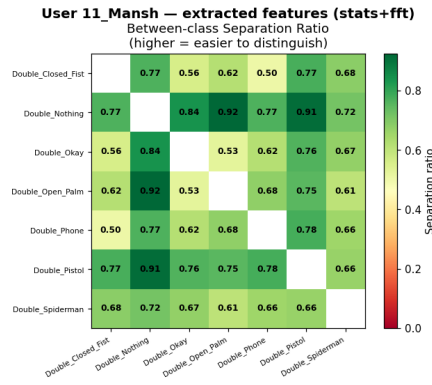
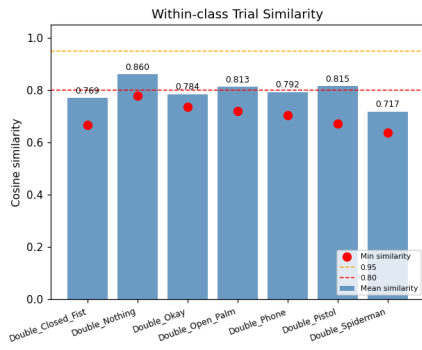
User 6_Harry — raw filtered trials



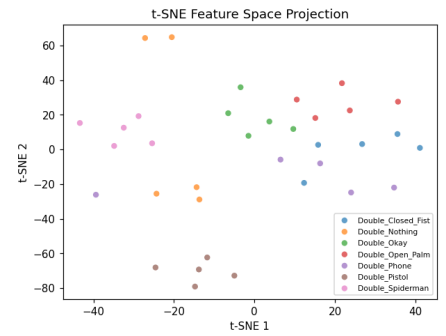
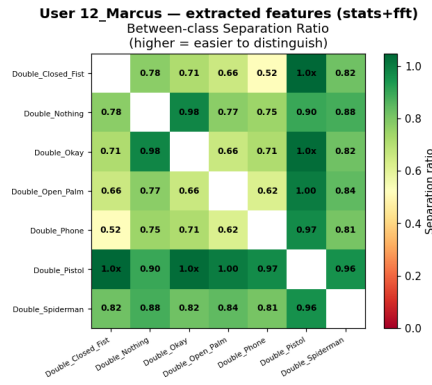
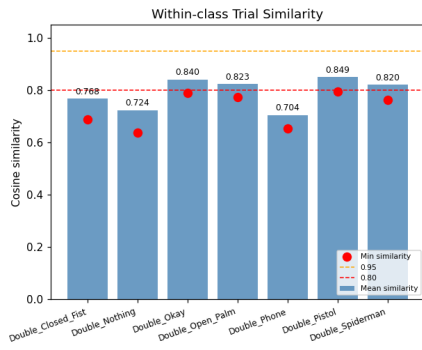
User 8_Jestin — raw filtered trials



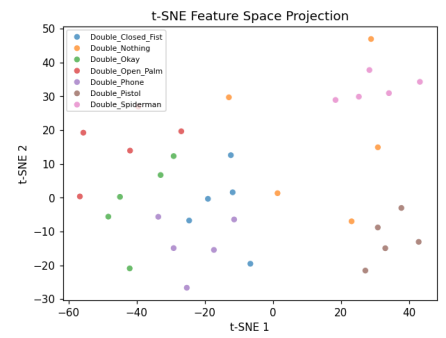
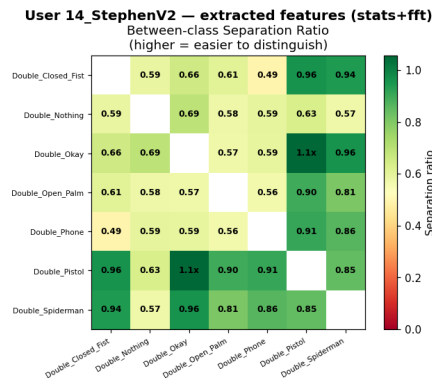
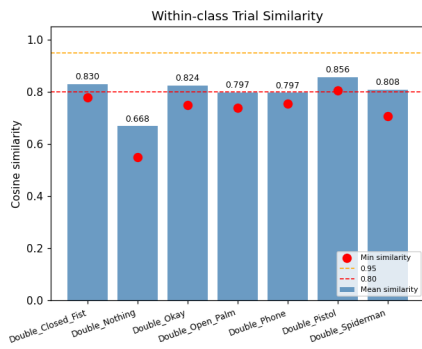
User 11_Mansh — extracted features (stats+fft)



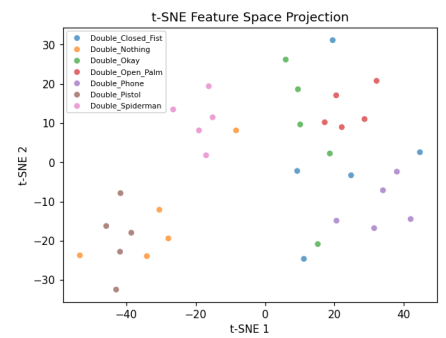
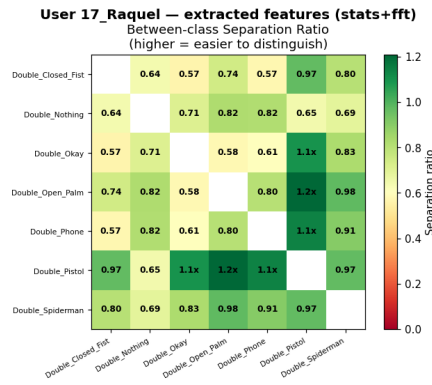
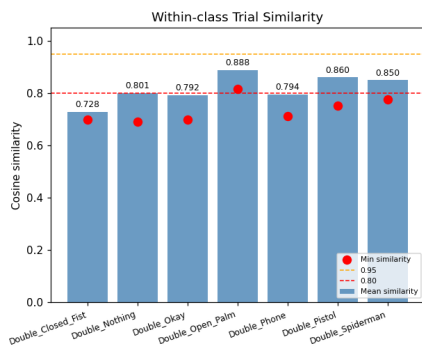
User 12_Marcus — extracted features (stats+fft)



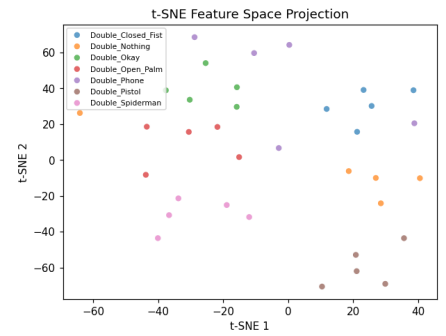
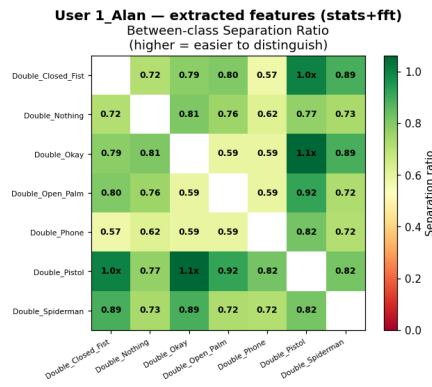
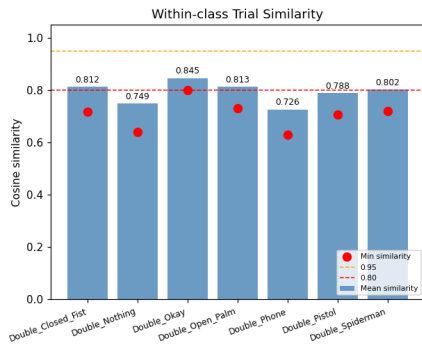
User 14_StephenV2 — extracted features (stats+fft)



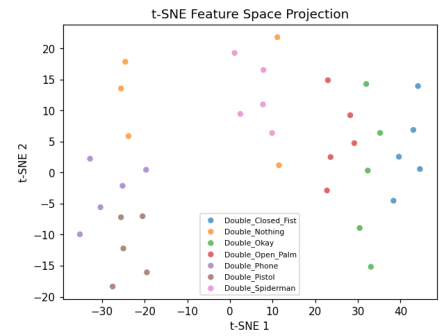
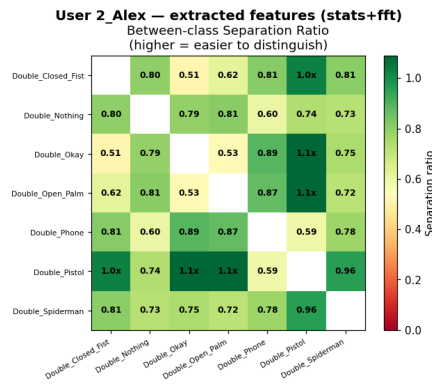
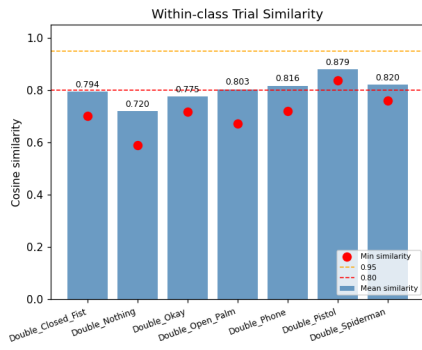
User 17_Raquel — extracted features (stats+fft)



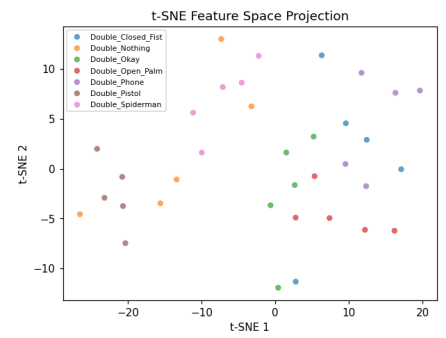
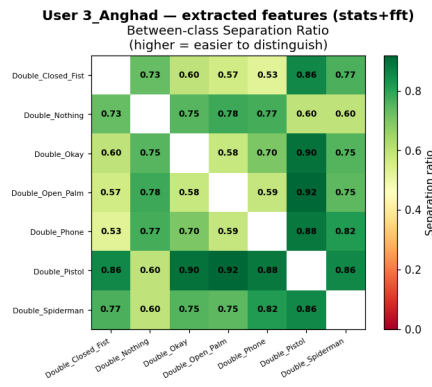
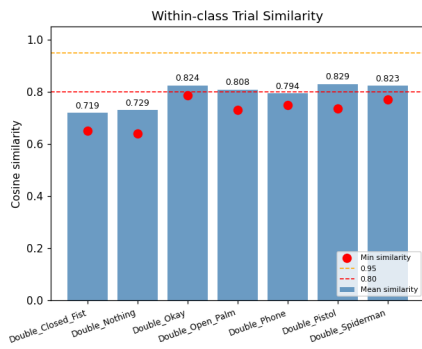
User 1_Alan — extracted features (stats+fft)



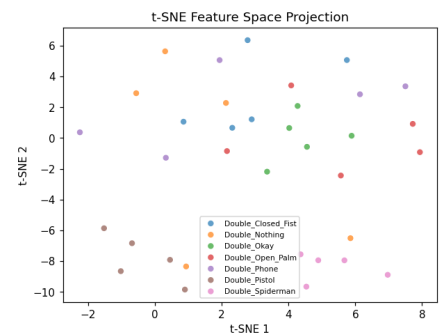
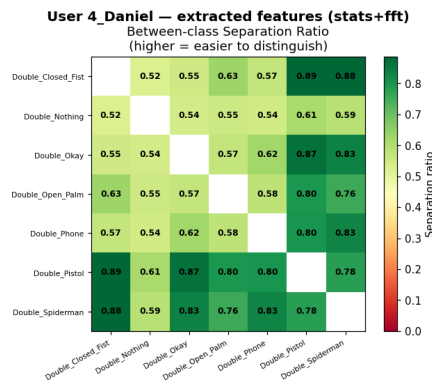
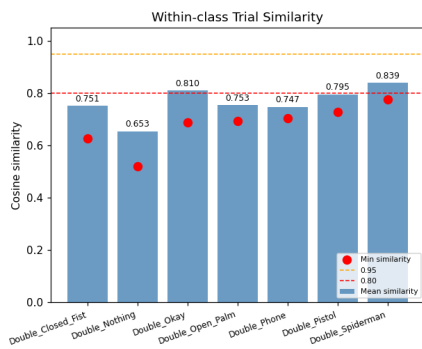
User 2_Alex — extracted features (stats+fft)



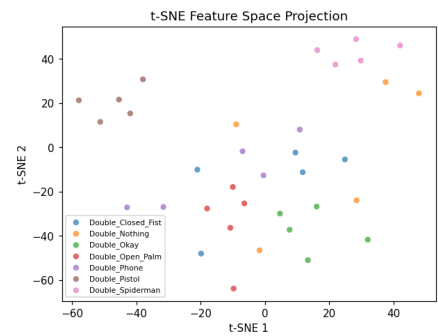
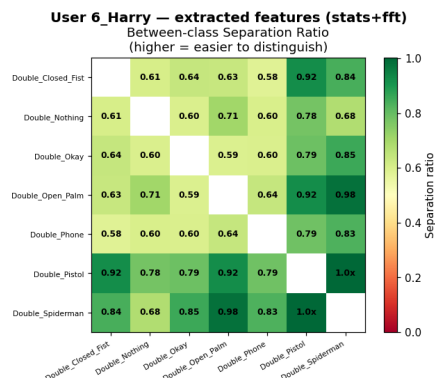
User 3_Anghad — extracted features (stats+fft)



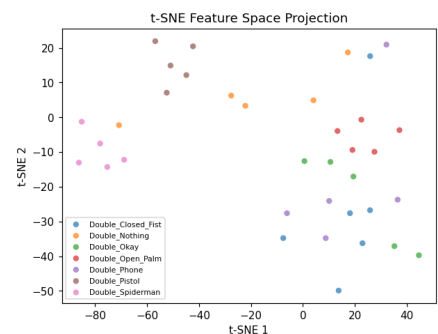
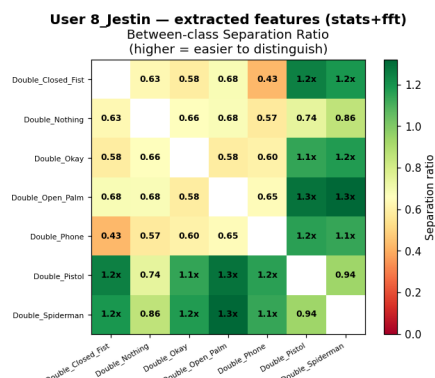
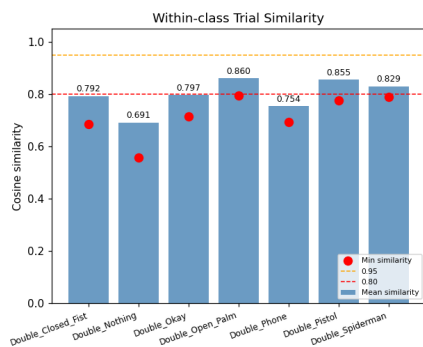
User 4_Daniel — extracted features (stats+fft)



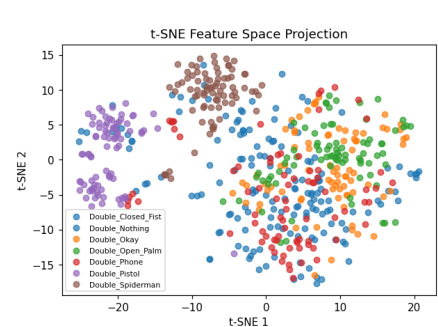
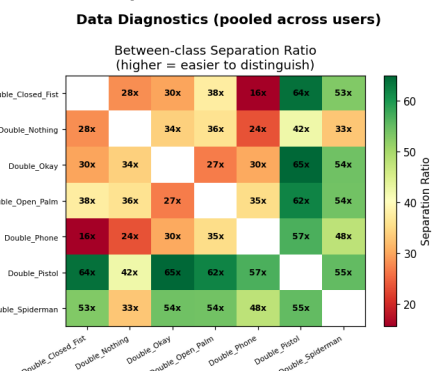
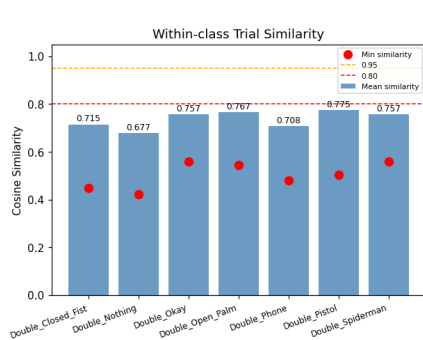
User 6_Harry — extracted features (stats+fft)



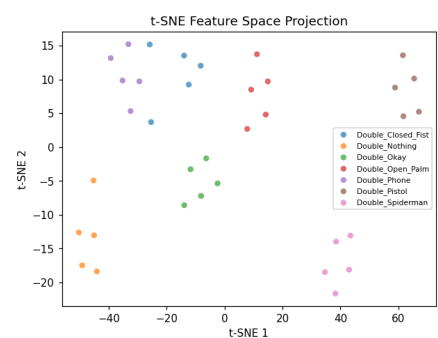
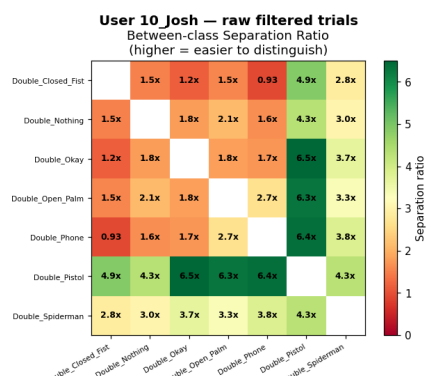
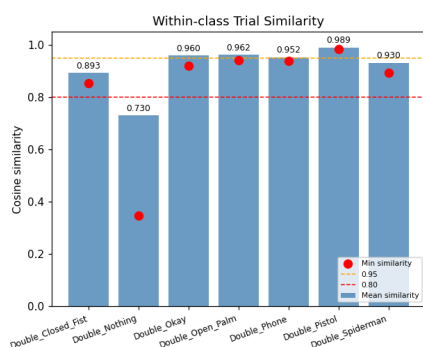
User 8_Jestin — extracted features (stats+fft)



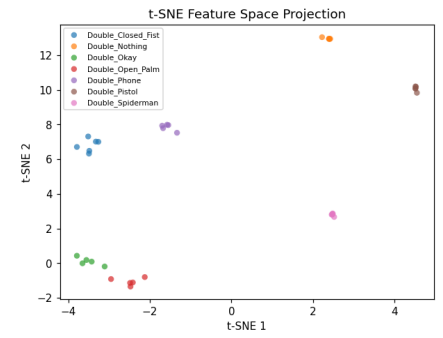
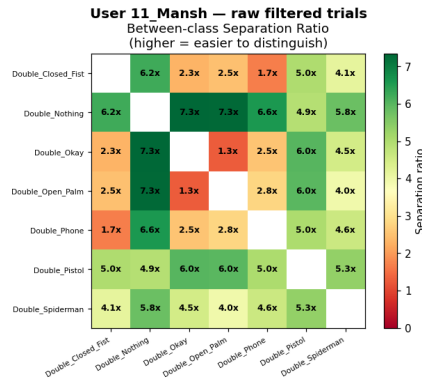
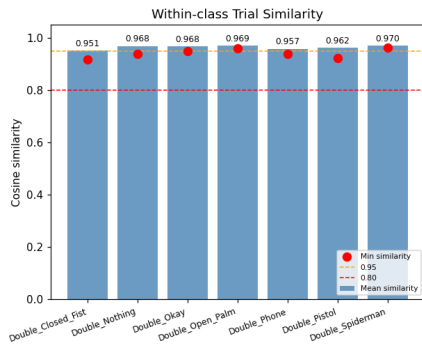
Pooled diagnostics — extracted features (across all users)



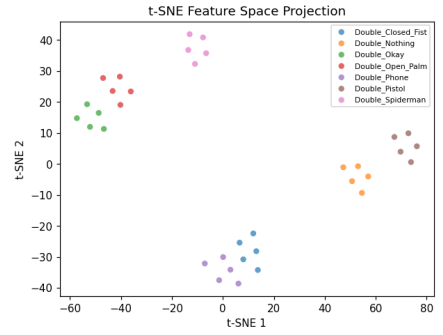
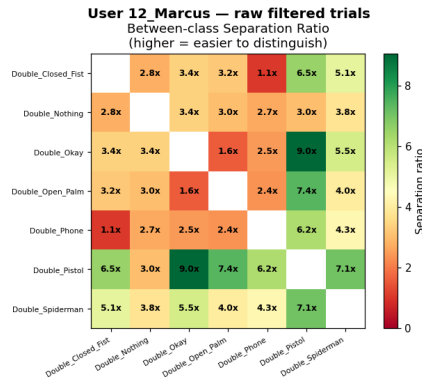
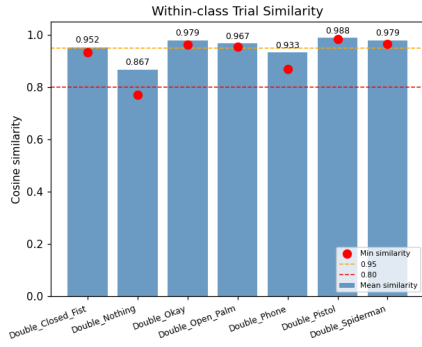
User 10_Josh — raw filtered trials



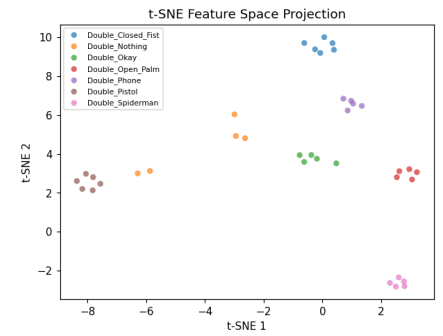
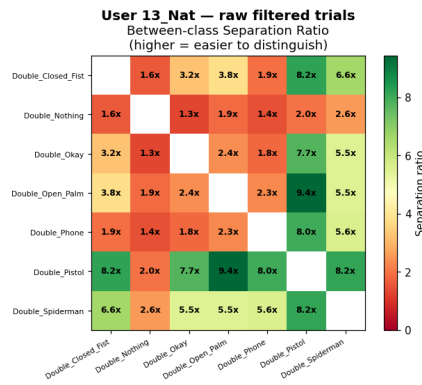
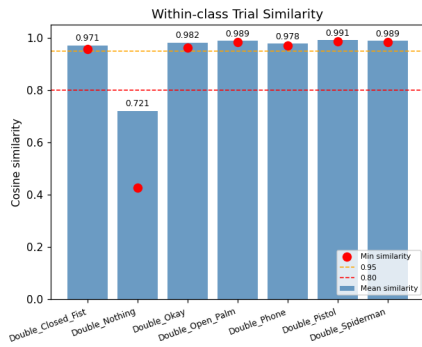
User 11_Mansh — raw filtered trials



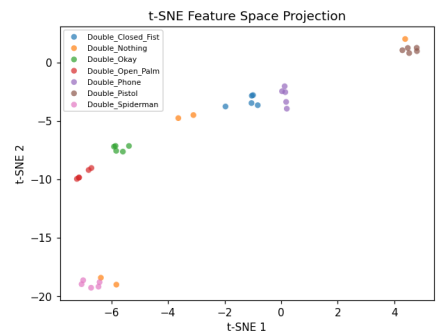
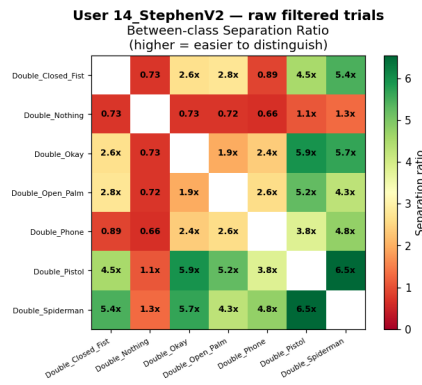
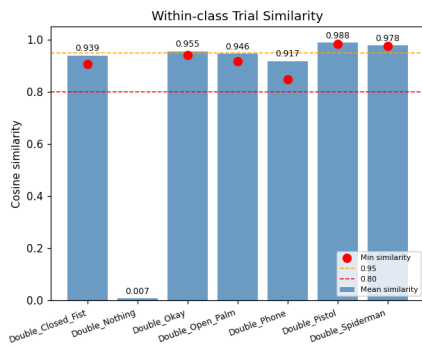
User 12_Marcus — raw filtered trials



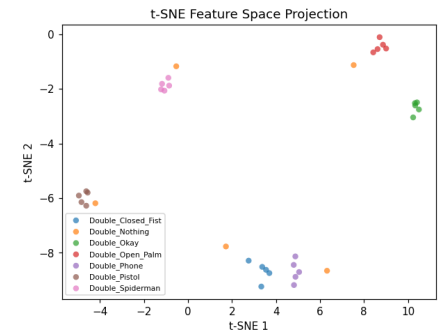
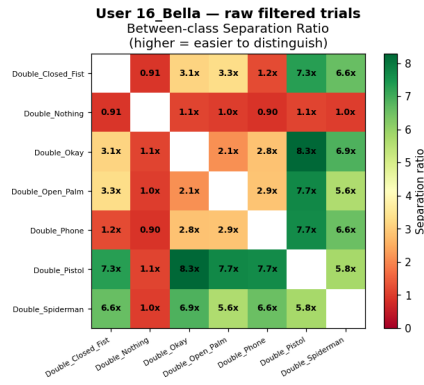
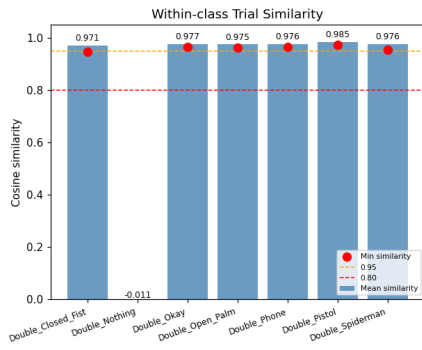
User 13_Nat — raw filtered trials



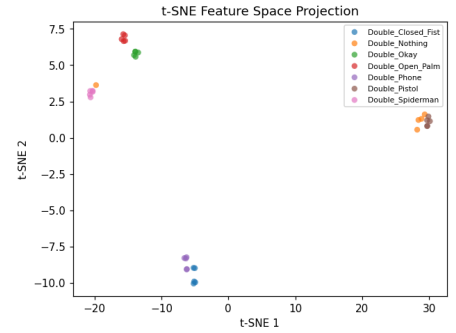
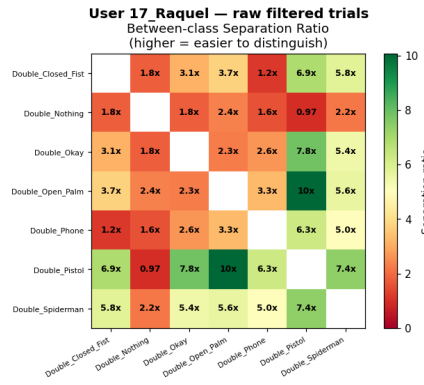
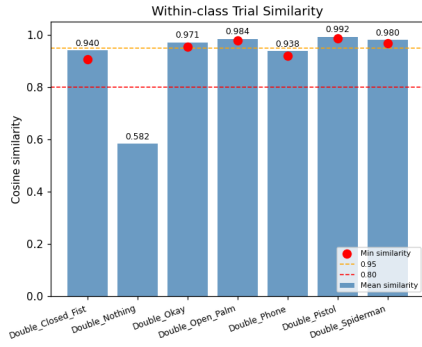
User 14_StephenV2 — raw filtered trials



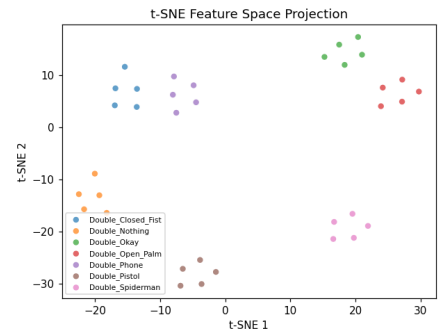
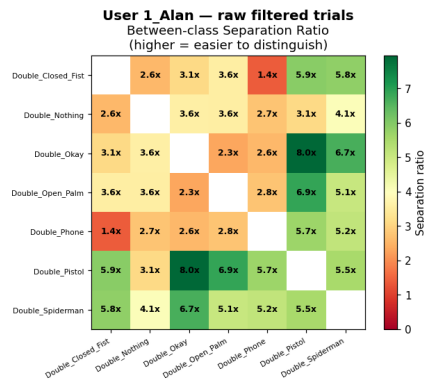
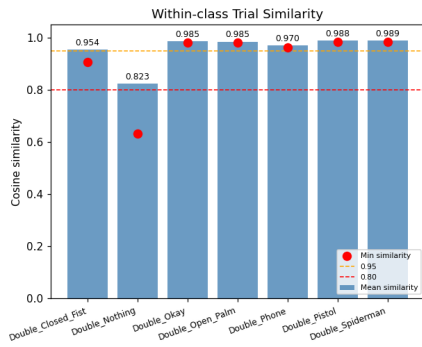
User 16_Bella — raw filtered trials



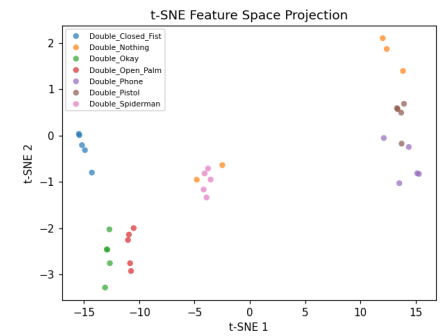
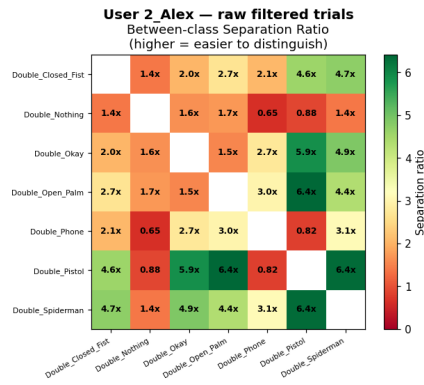
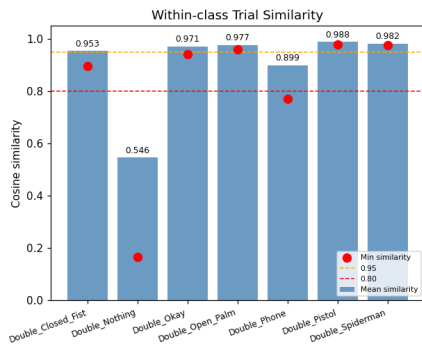
User 17_Raquel — raw filtered trials



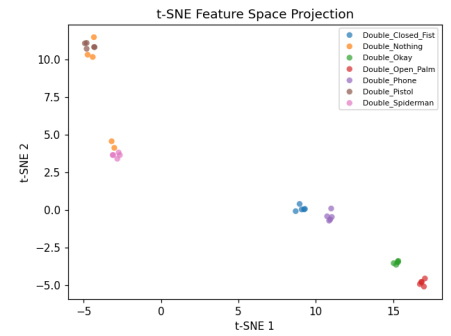
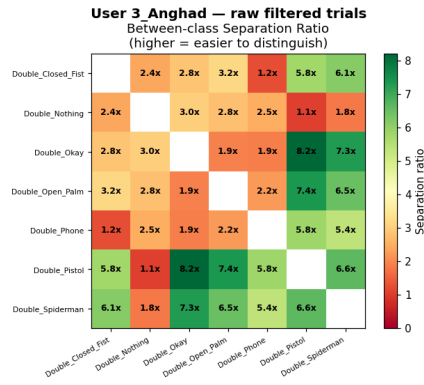
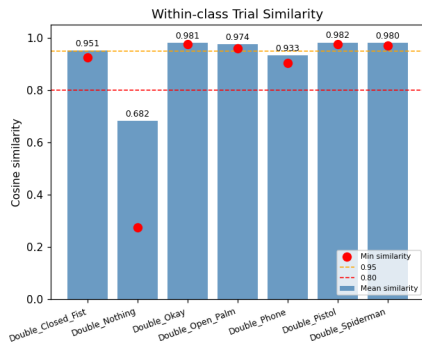
User 1_Alan — raw filtered trials



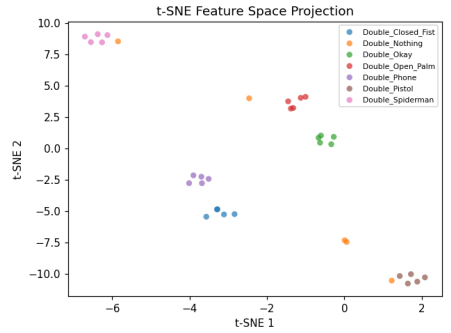
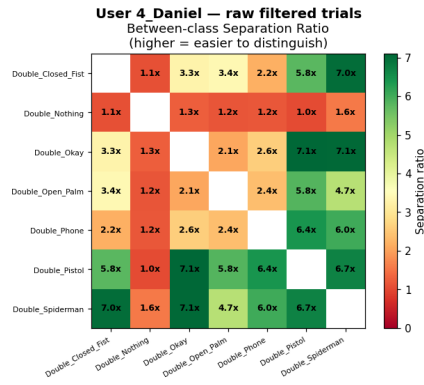
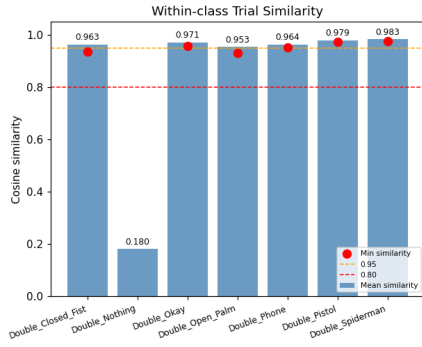
User 2_Alex — raw filtered trials



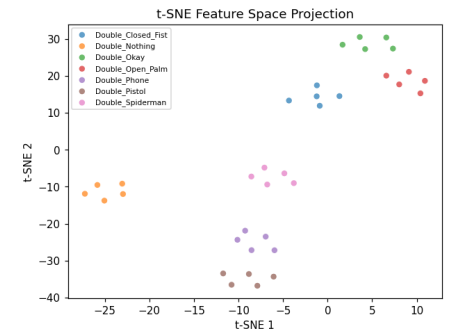
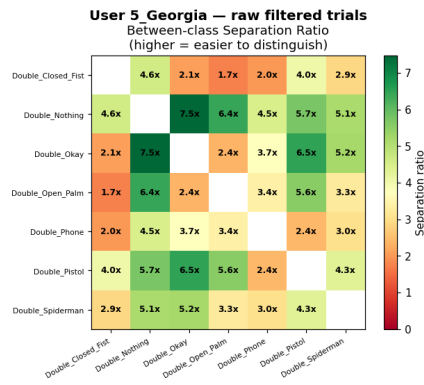
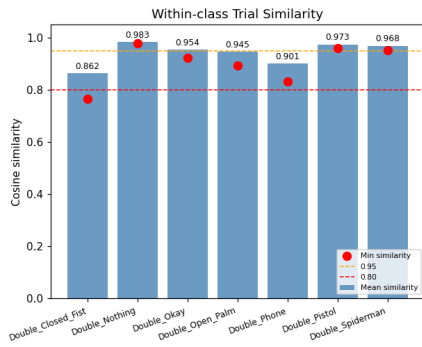
User 3_Angad — raw filtered trials



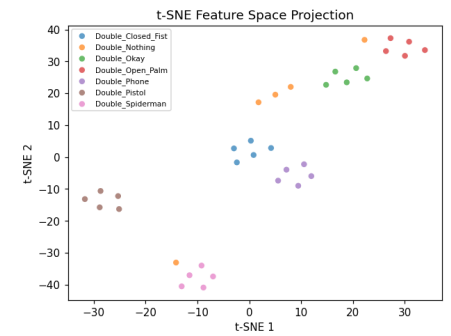
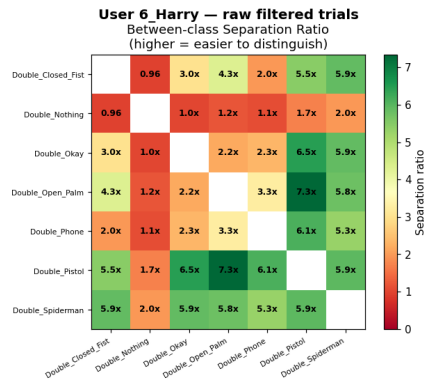
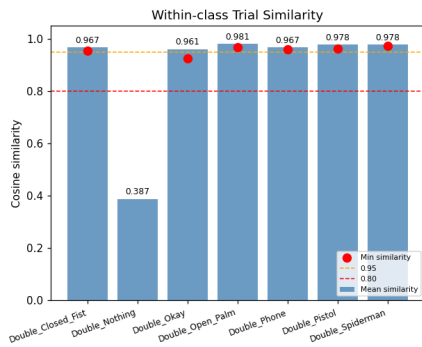
User 4_Daniel — raw filtered trials



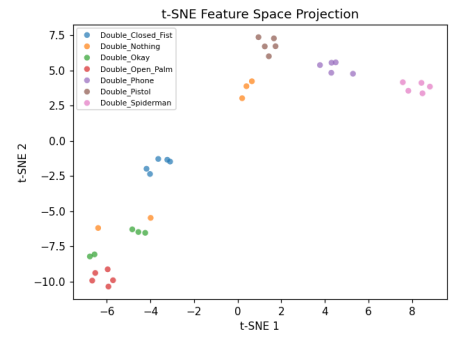
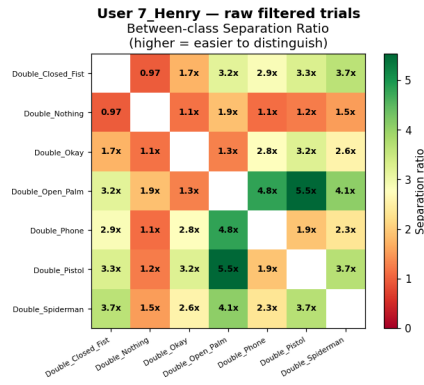
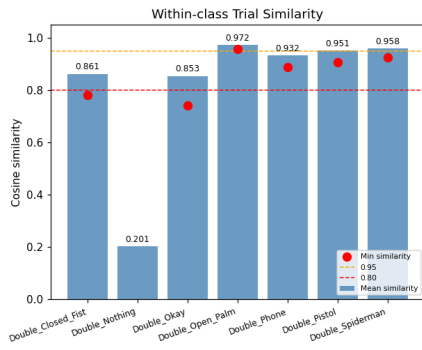
User 5_Georgia — raw filtered trials



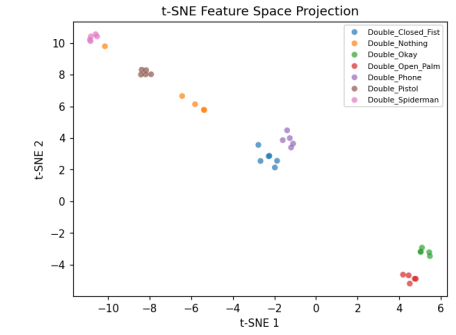
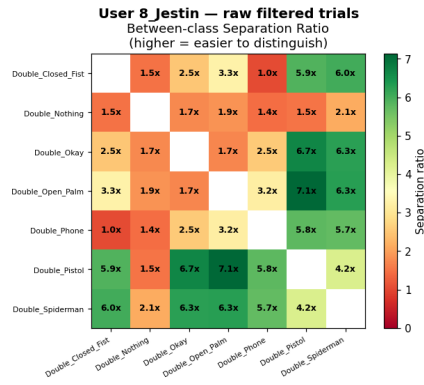
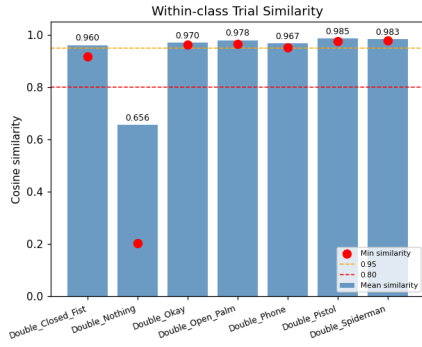
User 6_Harry — raw filtered trials



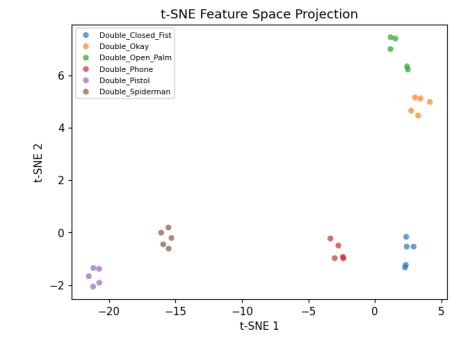
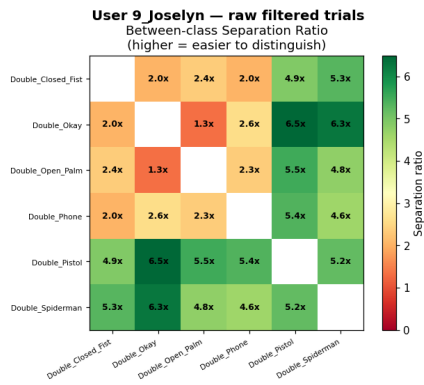
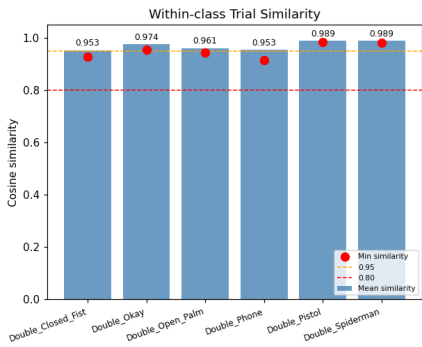
User 7_Henry — raw filtered trials



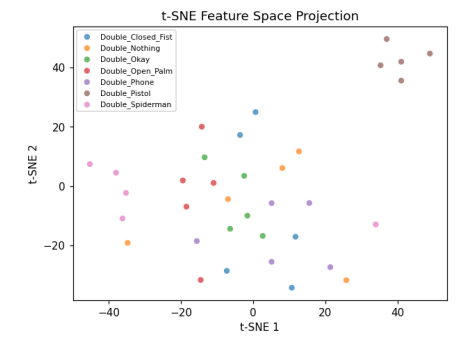
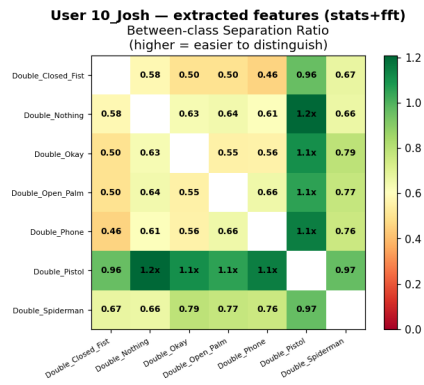
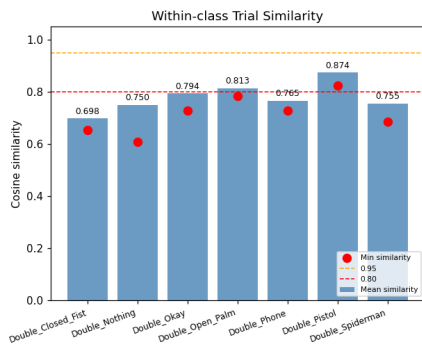
User 8_Jestin — raw filtered trials



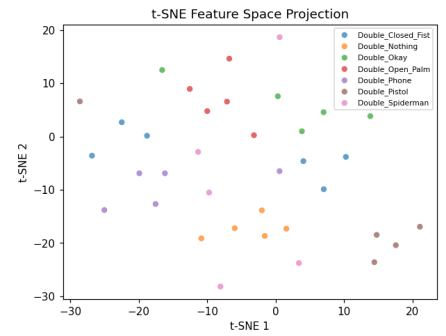
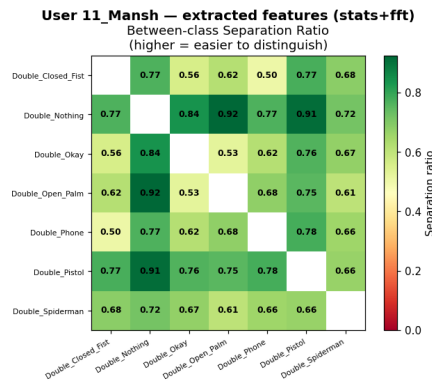
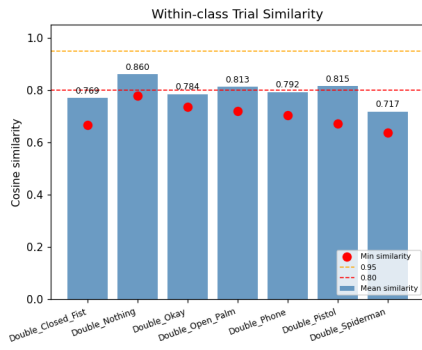
User 9_Joselyn — raw filtered trials



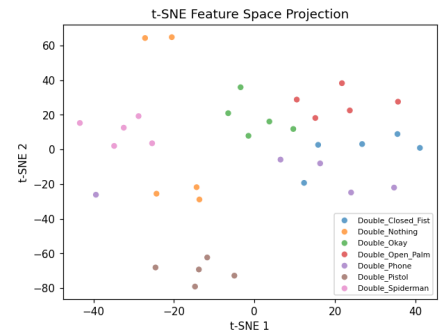
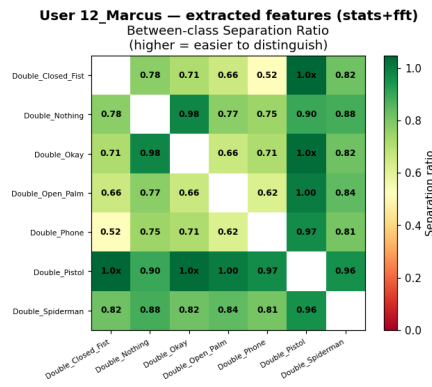
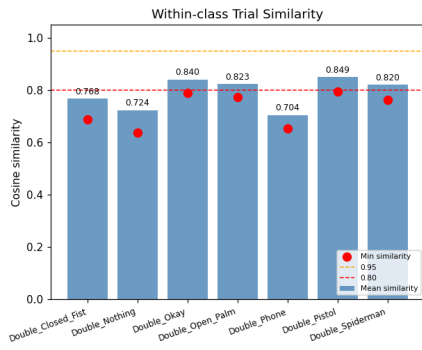
User 10_Josh — extracted features (stats+fft)



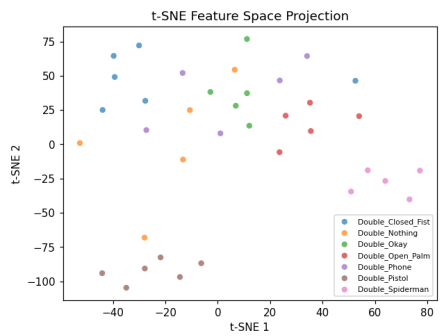
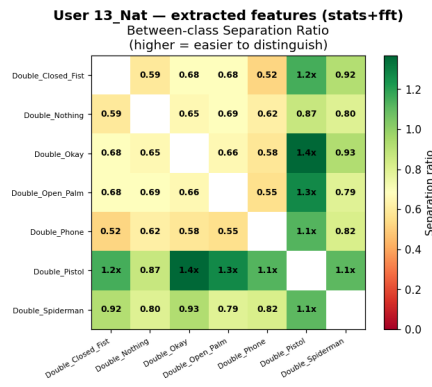
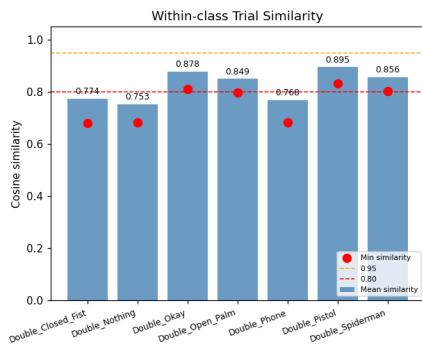
User 11_Mansh — extracted features (stats+fft)



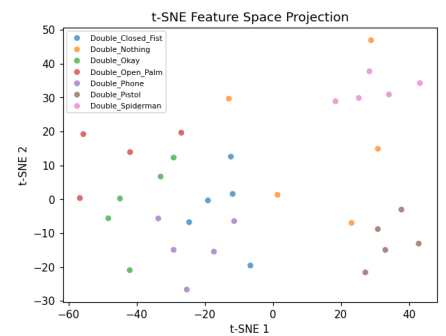
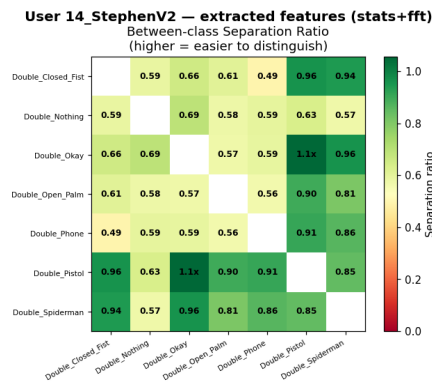
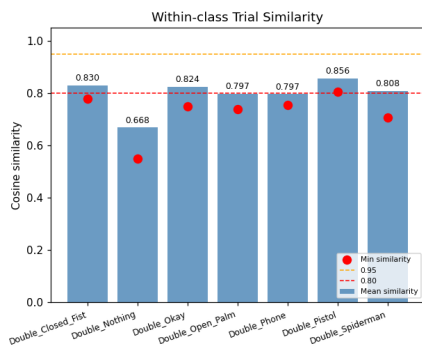
User 12_Marcus — extracted features (stats+fft)



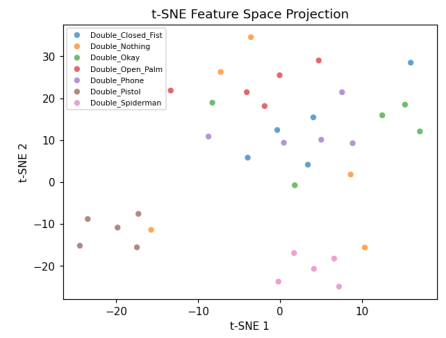
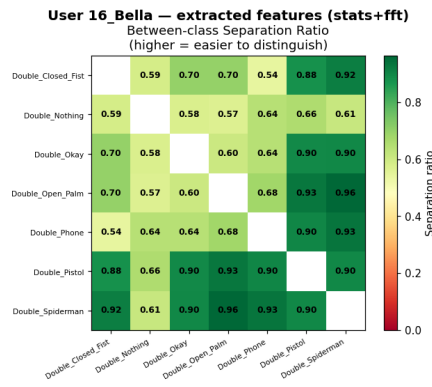
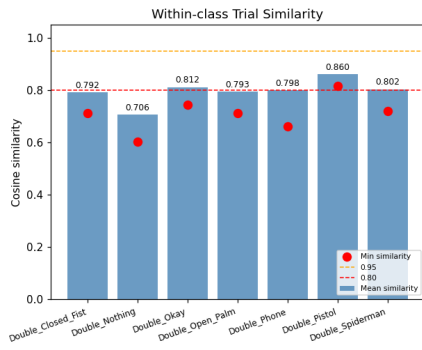
User 13_Nat — extracted features (stats+fft)



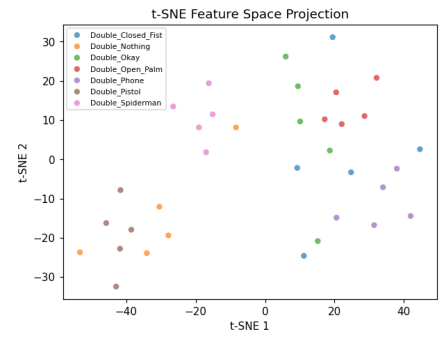
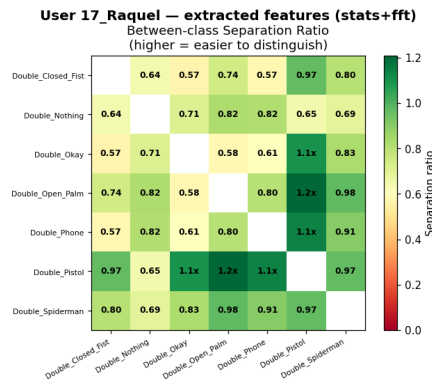
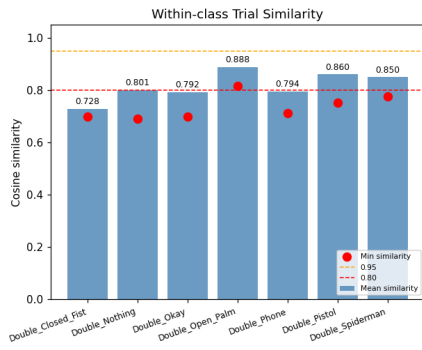
User 14_StephenV2 — extracted features (stats+fft)



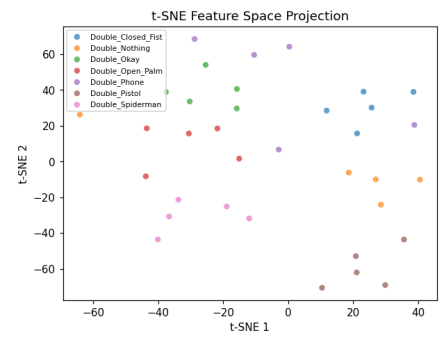
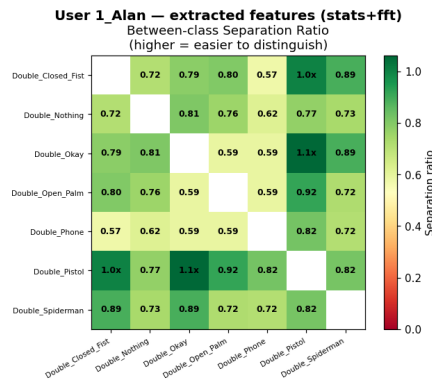
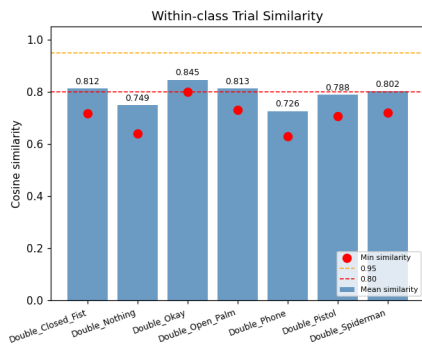
User 16_Bella — extracted features (stats+fft)



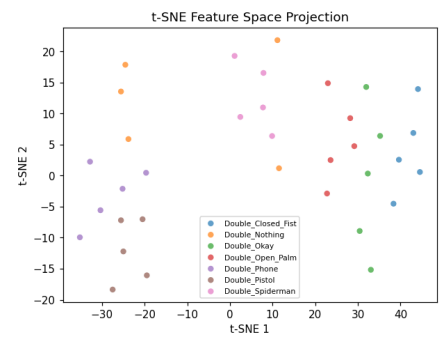
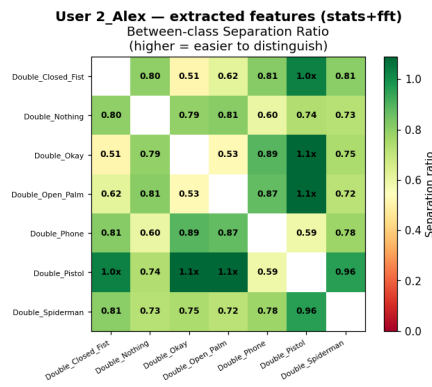
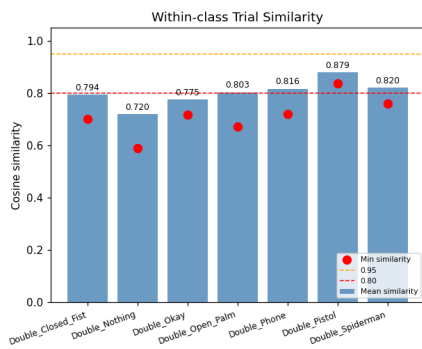
User 17_Raquel — extracted features (stats+fft)



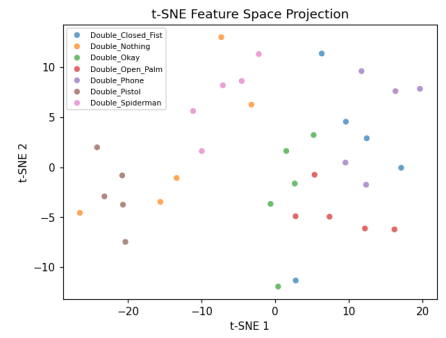
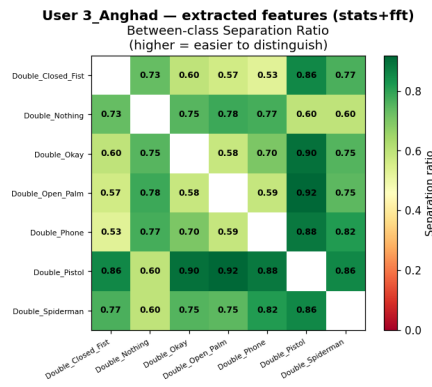
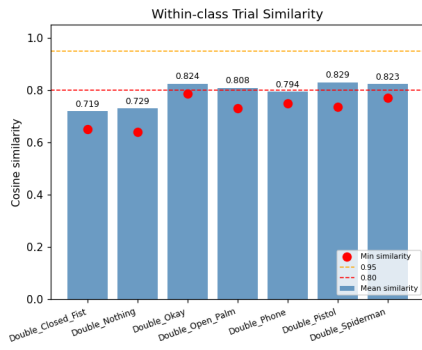
User 1_Alan — extracted features (stats+fft)



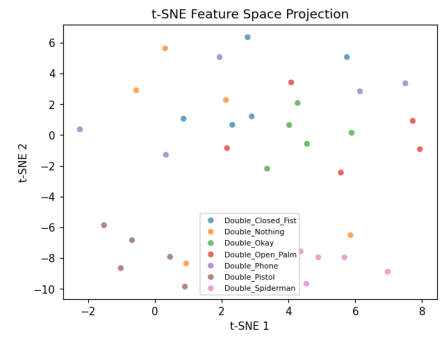
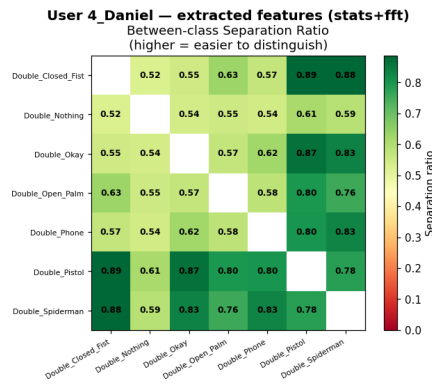
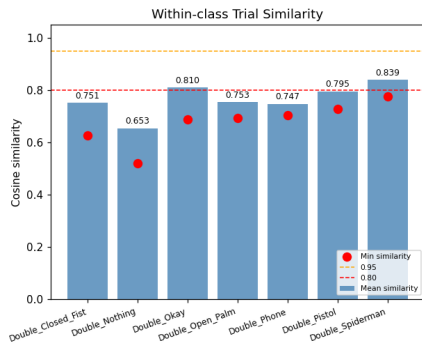
User 2_Alex — extracted features (stats+fft)



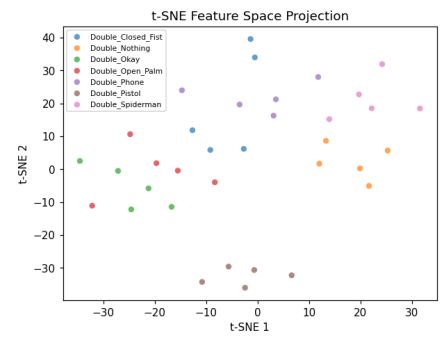
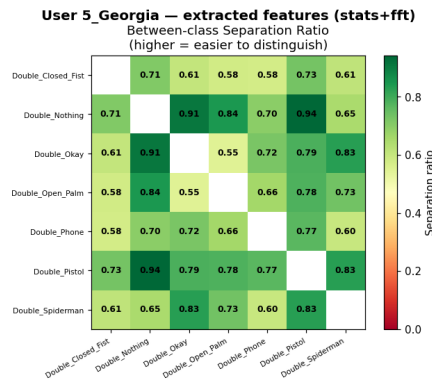
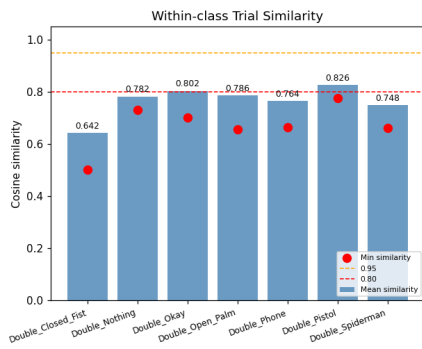
User 3_Angad — extracted features (stats+fft)



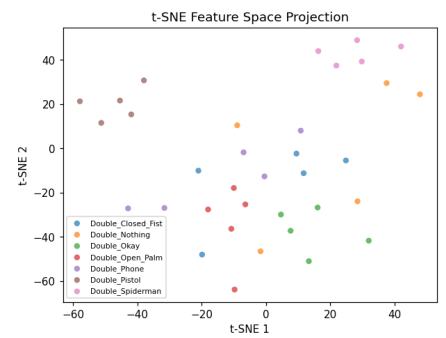
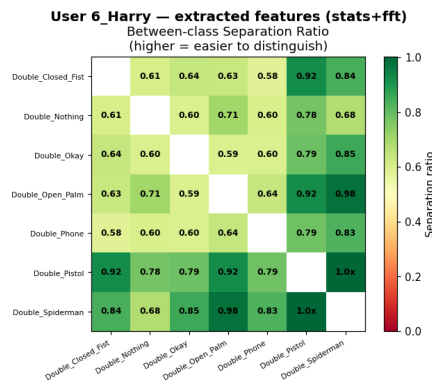
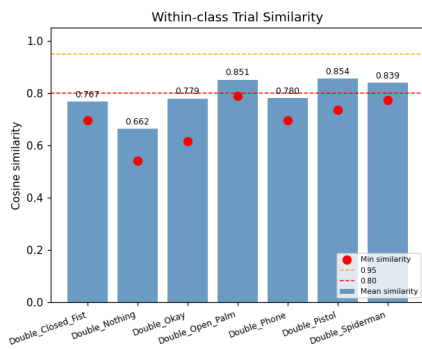
User 4_Daniel — extracted features (stats+fft)



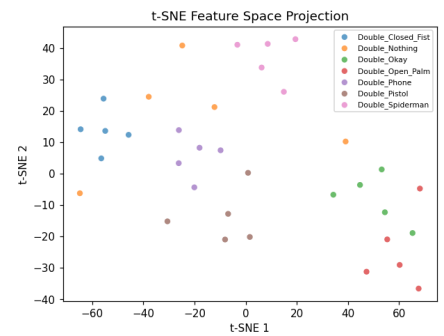
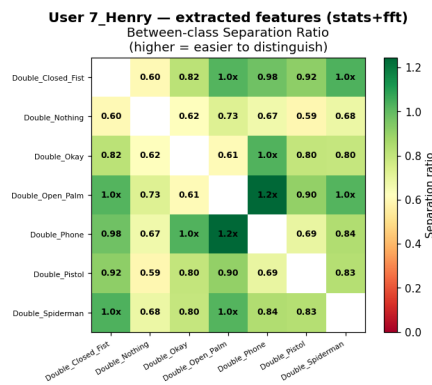
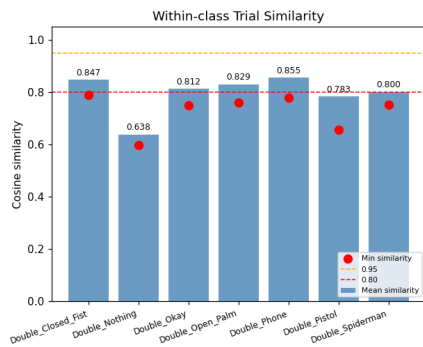
User 5_Georgia — extracted features (stats+fft)



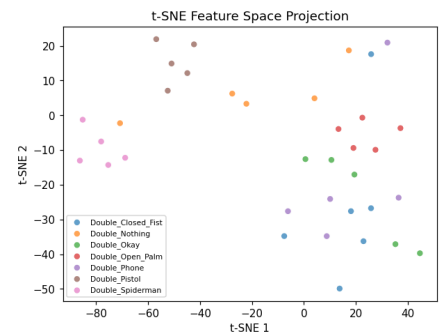
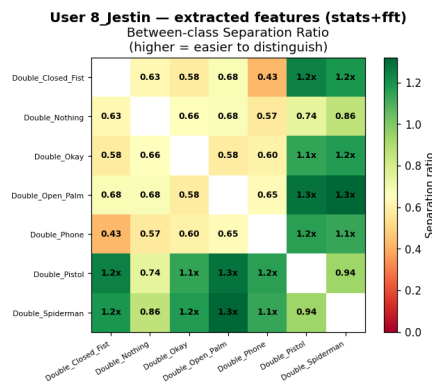
User 6_Harry — extracted features (stats+fft)



User 7_Henry — extracted features (stats+fft)



User 8_Jestin — extracted features (stats+fft)



User 9_Joselyn — extracted features (stats+fft)

